Reinforcer Postposing Revised:

Linearization and Exponence at the Syntax-Phonology Interface

Abstract: This paper investigates the nature of linearization and exponence by documenting and analyzing an output-optimizing alternation between long-distance displacement, suppression, and regular phonological repairs. Mandar (Austronesian, Central Indonesia) has a pair of monosyllabic locative adverbs that are routinely generated in DPs, forced to shift to the edge of the utterance, and suppressed when they cannot move. These steps are phonologically transparent in three respects: they ensure that these adverbs are only exponed at right edge of a phonological domain, they visibly resolve an output constraint, and they are suspended when the prosodic context enables a more optimal phonological repair. The paper thus argues that the systems of linearization and exponence must be split into two parts: a morphological stage that linearizes the terminal nodes of the syntax and matches them with vocabulary entries in a manner roughly familiar from Distributed Morphology (Halle & Marantz, 1993) and a phonological stage that reinstantiates this linear order and inserts the exponents that the morphology selects in the parallel, global, and output-optimizing calculus where prosodic structure is built (Selkirk, 2009, 2011).

Keywords:

Linearization, Exponence, Distributed Morphology, Match Theory, Prosodic Hierarchy

1 Introduction

It is a central target of minimalist inquiry to understand the structure of the PF branch: the series of feed-forward modules that translate syntactic representations into phonetic strings. In the domain of constituency, many lines of work have converged to suggest that representations undergo a sequential process of imperfect translation as they pass through the modules of this branch. The literature on Distributed Morphology (Halle & Marantz, 1993) for instance, has established that the terminal nodes of the syntax must be parsed into morphological words and subwords (Embick & Noyer, 2001) and then reworked through operations like node insertion, fission, fusion, and obliteration in an ordered and sequential morphology (Embick, 2000; Embick & Noyer, 2001; Arregi & Nevins, 2012; Hewett, 2023). The literature on the Prosodic Hierarchy, in turn, has established a similar network of results: morphological words and morphosyntactic phrases must be reinstantiated as prosodic words and phonological phrases in a parallel and global phonological calculus that reworks the ensuing phonological constituency in response to output constraints (Nespor & Vogel, 1986; McCarthy & Prince, 1993).

The goal of this paper is to advance our understanding of two systems that play out along this branch. The first is the system of Linearization, which establishes relations of linear precedence over the output correspondents of the unordered terminals of the syntax (Chomsky, 1995). The second is the system of Exponence, which associates these nodes with phonological content—in the terminology of Distributed Morphology, through a process of vocabulary insertion that pairs them with vocabulary entries from the lexicon (Halle & Marantz, 1993). The literature has established that certain parts of these systems must operate in the morphology: many patterns of allomorph selection are sensitive to information that is only available in that module (Halle & Marantz, 1993), many steps of movement seem to proceed over morphological representations

(Embick & Noyer, 2001; Embick, 2007; Arregi & Nevins, 2012; Myler, 2013), and many movements of this type feed allomorph selection (Adger, 2006; Ostrove, 2018). But there are reasons to believe that the phonology must play a role in each system as well: there are steps of movement that seem sensitive to prosodic constituency (Halpern, 1995; Bennett *et al.*, 2016; Aissen, 2017; Kusmer, 2019), patterns of exponence that reference global phonological context (McCarthy, 2011; Henderson, 2012; Brodkin, 2025b), and patterns of both types that seem to repair output constraints (Mester, 1994; Kager, 1996; Perlmutter, 1998; Booij, 1998; Rubach & Booij, 2001; Mascaró, 1996, 2007; Kurisu, 2001).

To refine the ensuing theoretical picture, this paper aims to document and analyze a system that shows optimizing alternations between displacement, non-exponence, and a regular phonological repair. This is one that governs the distribution of two locative adverbs in Mandar, an Austronesian language of Sulawesi. These adverbs are lexically selected by a subset of demonstratives and surface after many DPs at the edges of utterance-sized domains, along the lines in (1). In the terminology of Bernstein 1997, I will refer to these adverbs as REINFORCERS and their associated DPs as ASSOCIATES.

(1) *The Reinforcers*

a. De tomauweng *(**e**)
this grey one here
'This old dude *(here)'

b. Ka'bal i de' [$_{DP}$ **do** tomauweng **o**].
invincible 3ABS they say that grey one there
'That old dude there is invincible, they say.' Sikki *et al.* 1987, 541

The analytical puzzle emerges from three interlocking effects. When their associates surface in utterance-medial positions, first, the reinforcers are typically forced to shift to the right (2a). When the right edge is occupied by other reinforcers, second, they are typically not exponed (2b). Under specific phonological circumstances, third, both processes are suspended and the reinforcers begin to surface in utterance-medial positions after their associated DPs (2c). I will refer to these patterns of movement and suspended exponence as Reinforcer Postposing and Reinforcer Deletion.

(2) *Postposing and Deletion*

a. Naita i [$_{\text{DP}}$ **do** guru ___] [$_{\text{DP}}$ iting sanaeke] **o**.
   saw that teacher there that kid
   'That teacher there saw that kid.'

b. Naita i [$_{\text{DP}}$ **do** guru ___] [$_{\text{DP}}$ **de** sanaeke e].
   saw that teacher there this kid here
   'That teacher there saw this kid here.'

c. Mane naita i [$_{\text{DP}}$ **do** guru **o**] [$_{\text{DP}}$ de sanaeke e].
   just saw that teacher there this kid here
   'That teacher there just saw this kid here.'

The first task of the paper is to refine and extend the initial case, laid out in Brodkin 2026, that Reinforcer Postposing occurs in the phonology. I lay out three arguments for this analysis: it ignores the locality constraints of the syntax, it proceeds along a path that can only be defined in phonological terms, and it is rerouted by changes in prosodic constituency that can only be forced by output constraints. The combined force of these arguments is to suggest that it carries its targets to the edge of the largest domain of the Prosodic Hierarchy: the intonational phrase (Nespor & Vogel, 1986; Selkirk, 2009).

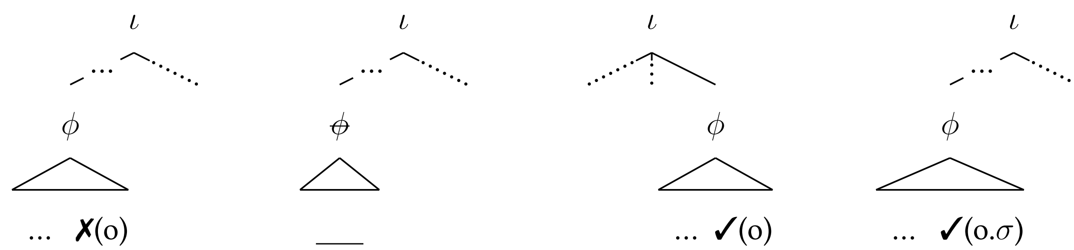
(3) *Phonological Displacement*

{$_{\iota}$ … *do* NP ___ … *o* }

The second task is to establish that Reinforcer Postposing and Reinforcer Deletion operate as item-specific repairs to a general output constraint. The case proceeds along the same logical path. First, I show that Reinforcer Postposing and Deletion are phonologically transparent: the reinforcers violate a constraint against phrase-final monosyllabic feet (4a) and the phonology resolves it by suspending exponence (4b) or moving them to positions that license these feet (4c). Second, I show that both processes are suspended when the driving constraint can be resolved in a different way (4d).

(4) *Output Optimization*

a. *Constraint* b. *Repair One* c. *Repair Two* d. *Suspension*



To capture these effects, I propose that linearization and exponence must each be split into two parts. In the morphology, I propose that the terminal nodes of the syntax are subjected to a first round of linearization and matched with vocabulary entries from the lexicon. In the phonology, I then propose that vocabulary entries are inserted into these terminal nodes and the linearization of the morphology is reimposed over their output correspondents in a characteristically imperfect way (and see also Bye & Svenonius 2012; De Belder 2020; Bennett *et al.* 2016; Kusmer 2019). The result is a framework that allows us to understand patterns of phonological relinearization and suspended exponence as cases of morphosyntax-phonology mismatch.

The remainder of this paper is organized as follows. Section 2 provides background information on Mandar and the reinforcers. Section 3 introduces Reinforcer Postposing and Reinforcer Deletion, partially reviewing the results of Brodkin 2026. Section 4 shows that both processes provide item-specific repairs to an output constraint. Section 5 develops an analysis of phonological relinarization and Section 6 develops an analysis of suspended exponence. Section 7 concludes with a cross-modular link to the theory of Greed (Chomsky, 1995; Bošković, 1995).

2 Background

Mandar is spoken by roughly 400,000 people in the Indonesian province of West Sulawesi. It is a language of the South Sulawesi subgroup, a primary subgroup of Malayo-Polynesian (Esser, 1938; Mills, 1975), and its closest relatives have been described in a small body of English-language work (Friberg, 1991, 1996; Strømme, 1994; Matti, 1994; Valkama, 1995a,b; Jukes, 2006; Lee, 2008; Kaufman, 2008; Laskowske, 2016; Finer, 1997, 1998, 1999; Béjar, 1999). This paper studies the standard variety of the language, which is spoken along the southern coastal plain of West Sulawesi (Grimes & Grimes, 1987) and which has been described in a grammar (Pelenkahu *et al.*, 1983) and discussed in many further publications by the Indonesian language office of South Sulawesi (Sikki *et al.*, 1987; Muthalib & Sangi, 1991; Friberg & Jerniati, 2000).

The data in this paper has been gathered through elicitation with a single speaker from the town of Ugibaru, Jupri Talib, with whom I have worked continually since 2018. The patterns of movement and deletion were encountered in 2019, documented in 2021, cross-checked in the field in 2022, and laid out in Brodkin 2026. The phonological systems that drive and suspend these effects, which are the focus of this paper, have come to light much more slowly through this time. Like the patterns of movement and deletion, however, they seem easy to replicate in many other dialects of Mandar and in several languages nearby, like Pattae'. I leave regional discussion to future work.

2.1 Clausal Syntax

Mandar is a head-marking and head-initial language that shows a consistent basic word order of VSO. It shows an ergative-absolutive system of agreement: transitive verbs agree with the S and second-position clitics agree with the transitive O and intransitive S.

The language requires postverbal arguments to surface before complement clauses and requires complement clauses to precede locative and temporal adjuncts to the *voice*P (5).[1]

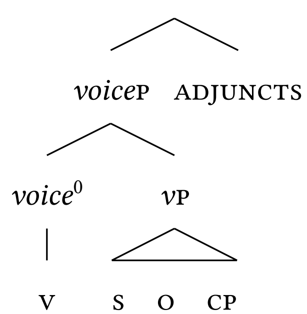
(5) *The Mandar Clause*

a. U-irrangngi i [$_S$ yau] [$_{CP}$ mua' lamba i marondong] dionging.
1ERG-hear 3ABS 1SG that go 3ABS tomorrow yesterday
'I heard yesterday that he was going tomorrow.'

b. U-pissang i [$_S$ yau] [$_O$ iKaco'] [$_{CP}$ mua' pole a'] marondong.
1ERG-tell 3ABS 1SG NAME that come 1ABS tomorrow
'I'll tell Kacho' that I'll come tomorrow.'

Brodkin 2025a shows that the VSO string forms a tight domain that corresponds to the extended VP. In this space, the verb consistently raises to a high head at the edge—by hypothesis, *voice*^{0}—and all following arguments and complement clauses are realized in their base positions in the *voice*P. The locative and temporal adjuncts that follow the VSO-CP string are adjoined to the *voice*P, yielding the rough clausal syntax in (6).

(6) *Mandar Clausal Syntax*



2.2 The Demonstrative-Reinforcer Construction

Mandar has four regular locative adverbs: *indi* 'right here', *dini* 'here', *dio* 'there', and *diting* 'over there'. It also has two monosyllabic locative adverbs that are synonymous with *dini* and *dio*: *e* 'here' and *o* 'there'. All of these elements can be independently adjoined to the *voice*P, as in (7a)-(7b), and the adverbs of the two sets can co-occur (7c).

(7) *The Locative Adverbs*

a. Urang i [**indi** / **dini** / **dio** / **diting**].
rain 3ABS right here here there over there
'It's raining [right here / here / there / over there].'

b. Urang i [**e** / **o**].
rain 3ABS here there
'It's raining [here / there].'

c. Mala bappa i [$_{voiceP}$ na-tarima akkatta-ta'] **diting** **o**.
can hopefully 3ABS 3ERG-accept purpose-2GEN over there there
'Hopefully they can accept your purpose there.' Friberg & Jerniati 2000, 243

The language also has four demonstratives that draw the same contrast in distance: *ndi* 'this' (very close), *de* 'this', *do* 'that', and *iting* 'that' (out of sight). Both types of locative adverbs can appear in the DPs that host these demonstratives to yield the demonstrative-reinforcer constructions below (Bernstein, 1997; Roehrs, 2010).

(8) *Demonstrative-Reinforcer Constructions*

a. Ndi posa indi e
this cat here here
'This cat here'

b. De posa dini e
this cat here here
'This cat here'

c. Do posa dio o
that cat there there
'That cat there'

d. Iting posa diting o
that cat there there
'That cat there'

In fragments of this type, the adverbs *e* and *o* are forced to appear after every DP that contains one of the demonstratives *ndi*, *de*, or *do*. The adverb *e* is required after *ndi* and *de* and the adverb *o* is required after *do*. In each case, it is ungrammatical to drop the adverb or replace it with anything else: *e* cannot be replaced with *o*, *o* cannot be replaced with *e*, and neither can be replaced with any of the regular locative adverbs. [2]

(9) *Selectional Requirements*

a. **De** posa *(**e**)
this cat here
'This cat *(here)'

b. **Do** posa *(**o**)
that cat there
'That cat *(there)'

These requirements are specific to *ndi*, *de*, and *do*. Neither adverb is ever required beneath *iting* or any D^0-like element in the language, such as the proprial article *i*.

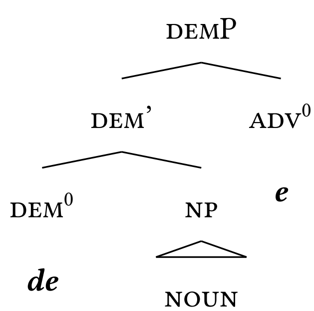
(10) *Lexical Idiosyncracy*

a. [$_{DP}$ **Iting** posa (o)].
that cat there
'That cat (there).'

b. [$_{DP}$ **i** Kaco' (e)].
D NAME here
'Kacho' (here).'

Brodkin 2026 thus proposes that *e* and *o* are LEXICALLY SELECTED by the demonstratives *ndi, de,* and *do*: these demonstratives spell out a head DEM^0 that systematically introduce *e* and *o* in their specifiers. In this context, I will refer to *e* and *o* as REINFORCERS and the DPs that contain them as their ASSOCIATES (Bernstein, 1997).

(11) *The Syntax of Selection*



2.3 Separation

Outside of fragments, the reinforcers surface in just the same way when their associates fall at the right edges of matrix clauses (12a). When their associates take clause-medial positions, however, we typically see the effect in (12b): outside of a configuration that depends on the distribution of prosodic words across the clause and the metrical status of the syllable that follows the associate (Section 4), they shift on their own to the right. I will refer to this process as REINFORCER POSTPOSING below.

(12) *Movement to the Right*

a.

Na-saka	i	[$_{\text{DP}}$	do	posa	**o**	].
3ERG-catch	3ABS		that	cat	there	

'That cat there caught it.'

b.

Na-saka	i	[$_{\text{DP}}$	do	posa	___	]	balao	**o**.
3ERG-catch	3ABS		that	cat			mouse	there

'That cat there caught the mouse.'

Reinforcer Postposing invariably places the reinforcers at the right edge of a clause-like domain. To this end, it routinely draws them across arguments (13a), adjuncts (13b), coordinate structures (13c), and complement CPs (13d).

(13) *Crossing Arguments and Adjuncts*

a.

Mane	na-pande	i	[$_{\text{DP}}$	do	Kaco'	___]	bau	posa	**o**.
just	3ERG-feed	3ABS		that	NAME		fish	cat	there

'That Kacho' there just fed the cat a fish.'

b.

Mam-buno	i	[$_{\text{DP}}$	de	posa	___]	ulo	kaiang	dio	di	bongi	**e**.
ANTIP-kill	3ABS		that	cat		snake	big	there	at	night	here

'This cat here killed a big snake there last night.'

c.

Na-peroa	i	[$_{\text{DP}}$	do	tau	___]	[$_{\text{CONJP}}$	iKaco'	na	iCicci'	]	**o**.
3ERG-invite	3ABS		that	person			NAME	and	NAME		there

'That person there invited Kacho and Chichi'.'

d.

Ma'ua	i	[$_{\text{DP}}$	do	Kaco'	___]	[$_{\text{CP}}$	mua'	macoa	i	mua'	lamba	a']	**o**.
said			that	NAME			that	good	if		go	1ABS	there

'That Kacho' there said that it would be good if I went.'

3 Phonological Displacement

The simplest analysis of this effect would treat it as a type of syntactic movement to the right, on a par with the Rightward Focus Movement in (14): one that targets certain types of foci and places them in positions linearized to the right of the *voice*P.

(14) *Rightward Focus Movement*

a. [$_{\text{FocP}}$ [$_{voice\text{P}}$ U-baca i ___] dionging [$_{\text{F}}$ BUKU-NNA ICICCI']].
1ERG-read 3ABS yesterday book-3GEN NAME
'I read CHICHI's BOOK yesterday.'

b. [$_{\text{FocP}}$ [$_{voice\text{P}}$ U-baca i [$_{\text{DP}}$ [$_{\text{D'}}$ buku-nna] ___]] dionging ICICCI'].
1ERG-read 3ABS book-3GEN yesterday NAME
'I read CHICHI's book yesterday.'

This section shows that Reinforcer Postposing must instead occur in the phonology, drawing the reinforcers across phonological representations to positions defined in surface prosodic terms. The case proceeds in three parts (and see Brodkin 2026 for partially overlapping arguments to this end). First, the reinforcers seem to move at a stage that lacks syntactic information, allowing them to escape islands and ignore c-command. Second, the reinforcers seem to move at a stage that has access to prosodic structure—suggesting that they must move after the morphosyntax. Third and finally, they seem to react to phonologically induced changes in prosodic constituency, suggesting that they shift in the phonological calculus where prosodic structure is built.

3.1 Island Effects

The work begins in the arena of absolute locality. Mandar shows a standard range of island constraints that prohibit extraction from many syntactic domains: for instance, from coordinate structures (15a) and complex NPs (15b). Rightward Focus Movement cannot escape from any islands of this type.

(15) *Syntactic Islands*

a. U-ita i [$_{\&P}$ sola-na iKaco' na sola-na iCicci'] allo ajuma'.
1ERG-see 3ABS friend-3GEN NAME and friend-3GEN NAME friday
'I saw Kacho's friend and Chichi's friend on Friday.'

b. Ma'ita a' [$_{NP}$ karewa [$_{CP}$ mua' pole i sola-na iCicci']] dio di FB.
I saw news that come 3ABS friend-3GEN NAME there on FB
'I saw [$_{NP}$ news [$_{CP}$ that Chichi's friend is coming]] on facebook.'

The reinforcers can escape all of these domains: for instance, the conjuncts of coordinate structures (16a)-(16b) and clauses embedded in NPs (16c).

(16) *Reinforcer Postposing: Escapes Islands*

a. U-ita i [$_{\&P}$ sola-na iKaco' na do tau ___] allo ajuma' **o**.
1ERG-see 3ABS friend-3GEN NAME and that guy friday there
'I saw Kaco's friend and that guy there on Friday.'

b. U-ita i [$_{\&P}$ do tau ___ na sola-na iKaco'] allo ajuma' **o**.
1ERG-see 3ABS that guy and friend-3GEN NAME friday there
'I saw that guy there and Kacho's friend on Friday.'

c. Ma'ita a' [$_{DP}$ karewa [$_{CP}$ mua' pole i do tau ___]] dio di FB **o**.
I saw news that come 3ABS that guy there on FB there
'I saw [$_{DP}$ news [$_{CP}$ that that guy there is coming]] on facebook.'

They also evade the Right Roof Constraint, which prevents rightward movement from embedded CPs from crossing elements in the matrix clause (Ross 1967). When embedded clauses are followed by right-adjuncts to the matrix *voice*P, then, they form islands for Rightward Focus Movement (17a) but not for Reinforcer Postposing (17b).

(17) *Reinforcer Postposing: Crosses Right Rooves*

a. *Pepeissang i [$_{CP}$ mua' pole i sola-na ___] **dio di facebook** ICICCI'.
Go check if come 3ABS friend-3GEN there on facebook NAME
INTENDED: 'Go check [$_{CP}$ if CHICHI's friend is coming] on facebook.'

b. Pepeissang i [$_{\text{CP}}$ mua' pole i do tau ___] **dio di facebook o**.
Go check if come 3ABS that guy there on facebook there
'Go check [$_{\text{CP}}$ if that guy there is coming] on facebook.'

3.2 Relative Locality

These effects are matched by a parallel deviation in the domain of Relative Locality. When multiple reinforcers are generated in a single domain, they surface after their associates when they can remain in situ. As a result, there are configurations where reinforcers surface after multiple arguments in the *voice*P (18a) or multiple constituents in the DP, like the head noun and a following possessor in SPEC,DP (18b).

(18) *Single Domains: Multiple Reinforcers*

a. Mane na-bengang i [$_{\text{DP}}$ de guru **e**] buku-na [$_{\text{DP}}$ do to basa **o**].
just 3ERG-give 3ABS this teacher here book-3GEN that linguist there
'This teacher here just gave his book to that linguist.'

b. [$_{\text{DP}}$ De buku-na **e** [$_{\text{DP}}$ guru-na [$_{\text{DP}}$ do sanaeke maroca' sannal **o**]]]
this book-3GEN here teacher-3GEN that kid loud very
'This book here of the teacher of those really loud kids.'

When multiple reinforcers are generated in positions that force movement, however, we see the following pair of effects. First, Reinforcer Postposing applies to the reinforcers that would otherwise surface linearly farthest to the right: those generated in the most deeply embedded rightward specifiers in the DP (19a) and the rightmost arguments in the *voice*P (19b). Second, all other reinforcers are forced to delete.

(19) *Reinforcer Competition*

a. Na-bengang i [$_{\text{DP}}$ de guru ___] buku-na [$_{\text{DP}}$ do to basa ___] digena' **o**.
3ERG-give 3ABS this teacher book-3GEN that linguist earlier there
'This teacher (here) gave his book to those loud kids earlier.'

b. [$_{\text{DP}}$ De buku-na ___ [$_{\text{DP}}$ guru-na [$_{\text{DP}}$ do ana' ___]] dio di meja] **o**.
this book-3GEN teacher-3GEN that kid there on table there
'This book (here) of the teacher of those kids there that's on the table.'

The relative locality constraints that guide this step are sensitive to linear order alone. As a result, they routinely forego reinforcers in structurally higher positions for reinforcers generated in lower positions that should surface farther to the right. The following examples illustrate this effect in VS-PP-X strings where reinforcers are generated in both subjects and PPs: when the reinforcers cannot remain in situ, the reinforcers in the PPs shift and the reinforcers in the subjects are not exponed.

(20) *Reinforcer Competition: Suppression*

a. Sidego i [$_{DP}$ do putra ___] sola [$_{DP}$ ndi putri ___] di wongi **e**.
dance 3ABS that prince with this princess at night here
'That prince (there) danced with this princess here last night.'

b. Di-anna i [$_{DP}$ de kocci' ___] tama [$_{DP}$ do kota' ___] digena' **o**.
PASS-put 3ABS this key into that box earlier there
'This key (here) was placed in that box there earlier.'

3.3 Phonological Transparency

These facts converge with two further effects to suggest that Reinforcer Postposing and Reinforcer Deletion operate over the hierarchical structure of the phonology (Selkirk 1984, 1986; Nespor & Vogel 1986). Following Itô & Mester 2007, 2009, 2012, 2013, I take this constituent structure to be built from the narrow inventory of abstract and universal prosodic domains in (21): the metrical mora, syllable, and foot (Liberman & Prince, 1977; Hayes, 1981), the interface categories of the prosodic word and phonological phrase (McCarthy & Prince, 1993; Truckenbrodt, 1995, 1999, 2002), and the speech-act category of the intonational phrase (Myrberg, 2010, 2013; Hamlaoui & Szendrői, 2015, 2017).

(21) *The Prosodic Hierarchy*

μ	>	σ	>	FT	>	ω	>	ϕ	>	ι
MORA		SYLLABLE		FOOT		WORD		PHRASE		INTONATIONAL PHRASE

The first argument begins from the diagnostics for the Intonational Phrase. Mandar has a process of Nasal Assimilation that assimilates coda /ŋ/ to all following segments in place and then denasalizes it before all but */m n ɲ ŋ/* and */b d d͡ʒ g/* (Mills, 1975; Pater, 1999). The following examples show how this process works in the prosodic word, using the top line to show its surface effect and the second line to show underlying forms. In affixes that end in */ŋ/*, this segment always assimilates in place to following */b d d͡ʒ g/* (22a) and then denasalizes before */p t t͡ʃ k/* (22b). [3]

(22) *Nasal Assimilation*

a.

mambawa,	**min**daiʔ,	**taɲ**d͡ʒappoʔ,	**saŋ**gallas
maŋ-bawa	miŋ-daiʔ	taŋ-d͡ʒappoʔ	saŋ-gallas
ANTIP-bring	VERBALIZER-up	NEG-destroy	ONE-glass

'bring, ascend, undestroyed, one glass.'

b.

mappoloŋ,	**mit**tama,	**tat**t͡ʃappuʔ,	**sak**kat͡ʃa
maŋ-poloŋ	miŋ-tama	taŋ-t͡ʃappuʔ	saŋ-kat͡ʃa
ANTIP-cut	VERBALIZER-in	NEG-exhausted	ONE-cup

'cut, enter, unexhausted, one cup.'

This process applies between words in all the strings that are not split by any obvious pause: for instance, fragments (23a) and root clauses that contain strings of arguments (23b), coordinate structures (23c), and embedded CPs (23d) within the *voice*P.

(23) *Nasal Assimilation: Through the Utterance*

a.

{$_\iota$ teɹo**s**	**s**ola sapi**k**	**k**e**s**	**s**annal }
tedoŋ	sola sapiŋ	kaiaŋ	sanna[1]
water buffalo	and cow	big	very

'Very big water buffalo and cows.'

b.

{$_\iota$ pure	napolo**p**	**p**aʔ ramar	**r**oppo**t**	**t**eʔe }
pura i	na-poloŋ	puaʔ ramaŋ	roppoŋ	teʔe
done 3ABS	[$_V$ 3ERG-cut]	[$_S$ MR. NAME]	[$_O$ grass]	[$_X$ now]

'Mr. Ramang is finished cutting the grass now.'

c. {$_\iota$ malli iramat **t**eɹos sola sapi**t** ton djoloʔ }.
maŋ-alli i iramaŋ tedoŋ sola sapiŋ tauŋ dioloʔ
ANTIP-buy 3ABS [$_S$ NAME] [$_{CONJP}$ buffalo and cow] [$_X$ year last]
'Ramang bought buffalos and cows last year.'

d. {$_\iota$ wirraɲɲi maʔ pole iramam marondo**t** **t**on djoloʔ }.
u-irraɲɲi i muaʔ pole i iramaŋ marondoŋ tauŋ dioloʔ
1ERG-hear 3ABS [$_{CP}$ that come 3ABS NAME tomorrow] year last
'I heard that Ramang was coming tomorrow a year ago.'

Despite this fact, it is blocked at a series of junctures that host a type of pause that I will term Comma Intonation: one that has the same phonetic shape and surface distribution as the prosodic juncture marked as ToBi Break Index 4 in English. This type of pause appears systematically around four types of syntactic domains. The first are left-peripheral topics, which are always followed by Comma Intonation whether they are raised to the left periphery (24a) or base-generated in that domain and resumed with pronouns below (24b). At their right edges, in each case, Nasal Assimilation is blocked.

(24) *Left-Peripheral Topics*

a. {$_\iota$ irama**ŋ** }, {$_\iota$ **t**aʔwita i }.
iramaŋ taʔ-u-ita i
NAME NOT-1ERG-see 3ABS *t*
'Ramang, I didn't see.'

b. {$_\iota$ me rama**ŋ** }, {$_\iota$ **t**aʔwita i }.
muaʔ iramaŋ taʔ-u-ita i
as for NAME NOT-1ERG-see 3ABS *pro*
'As for Ramang, I didn't see him.'

The second are the right-peripheral topics in (25), which are base-generated in high positions, linearized after the *voice*P, and resumed by pronouns (at least when nominal) in the preceding clause. These topics form islands for Nasal Assimilation as well.

(25) *Right-Peripheral Topics*

a. {$_\iota$ nete iramaŋ }, {$_\iota$ **k**indoʔnaʔ }
na-ita i iramaŋ kindoʔ-na
3ERG-see 3ABS NAME *pro* mom-3GEN

'Ramang saw her, his mom.'

b. {$_\iota$ nete iramaŋ }, {$_\iota$ **t**on djoloʔ }
na-ita i iramaŋ tauŋ dioloʔ
3ERG-see 3ABS NAME year last

'Ramang saw her, last year.'

The third are appositives and parentheticals (26a) and the fourth are initial adjunct CPs, which behave like their counterparts in English: they can either be set off by Comma Intonation or not (26b)-(26c). When these constituents are offset, they are opaque to Nasal Assimilation; when initial adjunct CPs are not, they are transparent instead.

(26) *Appositives and Initial Adjunct CPs: Parallel Effects*

a. {$_\iota$ nete iramaŋ }, {$_\iota$ kapala ɹesa }, {$_\iota$ ton djoloʔ }.
na-ita i ramaŋ kapala desa tauŋ dioloʔ
3ERG-see 3ABS NAME head village year last

'She saw Ramang, the mayor, last year '

b. {$_\iota$ maʔ pole iramaŋ }, {$_\iota$ **p**ole toaʔ }.
muaʔ pole i iramaŋ pole to aʔ
if come 3ABS NAME come also 1ABS

'If Ramang's coming, I'm coming.'

c. {$_\iota$ maʔ pole iramap **p**ole toaʔ jau }.
muaʔ pole i iramaŋ pole to aʔ iau
if come 3ABS NAME come also 1ABS 1SG

'If Ramang's coming I'M ALSO coming.'

These facts suggest that Nasal Assimilation and Comma Intonation are keyed to the right edge of an abstract prosodic domain that is built around complete utterances (27a)-(27b) and a narrow set of syntactic domains: left- and right-peripheral topics

(27c)-(27e), appositives and parentheticals (27f), and most initial adjunct CPs (27g)-(27h). This is the Intonational Phrase.[4]

(27) *The Intonational Phrase*

a. {$_\iota$ My sister }.

b. {$_\iota$ I saw my sister }.

c. {$_\iota$ {$_\iota$ My sister }, {$_\iota$ I saw } }.

d. {$_\iota${$_\iota$ As for my sister }, {$_\iota$ I saw her }}.

e. {$_\iota$ {$_\iota$ I'd see her }, {$_\iota$ my sister } }.

f. {$_\iota$ {$_\iota$ I'd see Zoe }, {$_\iota$ my sister } }.

g. {$_\iota$ {$_\iota$ If my sister went }, {$_\iota$ I'd go } }.

h. {$_\iota$ If my sister went I'd GO }.

From this position, we can establish three layers of phonological structure around the process of Reinforcer Postposing. First, the reinforcers remain in-situ when their associates are final in the intonational phrase. As a result, they follow the associates that are left-peripheral topics (28a)-(28b), right-peripheral topics (28c), and appositives (28d).

(28) *Reinforcers: Follow Associates that are Final in the Intonational Phrase*

a. { do ramaŋ **oʔ** }, { wita i }.
do ramaŋ o u-ita i
that NAME there 1ERG-see 3ABS *t*
'That Ramang there, I saw.'

b. { maʔ do ramaŋ **oʔ** }, { wita i }.
muaʔ do ramaŋ o u-ita i
as for that NAME there 1ERG-see 3ABS *pro*
'As for that Ramang there, I didn't see him.'

c. {$_\iota$ nete solau ton djoloʔ }, {$_\iota$ do ramaŋ **oʔ** }.
na-ita i sola-u tauŋ dioloʔ do ramaŋ o
3ERG-see 3ABS *pro* friend-1GEN year last that NAME there
'He saw my friend last year, that Ramang there.'

d. {$_\iota$ nete kapala ɹesa }, {$_\iota$ do ramaŋ **oʔ** }, {$_\iota$ ton djoloʔ }.
na-ita i kapala desa do ramaŋ o tauŋ dioloʔ
3ERG-see 3ABS head village that NAME there year last
'She saw the mayor, that Ramang there, last year '

Second, Reinforcer Postposing always proceeds from positions that are medial in the intonational phrase and lands at its edge. As a result, it can place its targets at the edges of the strings that precede appositives (29a) and right-peripheral topics (29b) and the right edges of the adjunct CPs that form their own intonational phrases ((29c); *cf.* (29d)).

(29) *Reinforcers: Move to the Right Edge of the Intonational Phrase*

a. {ι nete ɹo ramas ___ solau **oʔ** }, {ι ton djoloʔ }.
na-ita i do ramaŋ sola-u o tauŋ dioloʔ
3ERG-see 3ABS that NAME friend-1GEN there year last
'That Ramang there saw my friend, last year.'

b. {ι nete ɹo tau ___ iramaŋ **oʔ** }, {ι kapala ɹesa }, {ι ton djoloʔ }.
na-ita i do tau iramaŋ o kapala desa tauŋ diolo
3ERG-see 3ABS that person NAME there head village year last
'That person there saw Ramang, the mayor, last year '

c. { maʔ pole ɹo ramam ___ marondoŋ **oʔ** }, {ι pole toaʔ }.
muaʔ pole i do ramaŋ marondoŋ o pole to aʔ
if come 3ABS that NAME tomorrow there come also 1ABS
'If that Ramang there is coming tomorrow, I'm coming too.'

d. { maʔ pole ɹo ramam ___ marondo**p** **p**ole toaʔ **oʔ** }.
muaʔ pole i do ramaŋ marondoŋ pole to aʔ o
if come 3ABS that NAME tomorrow come also 1ABS 1SG there
'If that Ramang there's coming tomorrow I'm ALSO coming.'

Third, Reinforcer Postposing always proceeds within the minimal intonational phrase. As a result, it cannot draw the reinforcers out of left-peripheral topics (30a)-(30b), appositives (30c), or prosodically independent initial adjunct CPs (30d).

(30) *Reinforcers: Trapped within the Intonational Phrase*

a. { do ramaŋ **oʔ** }, { wita i ___ }.
do ramaŋ o u-ita i
that NAME there 1ERG-see 3ABS *t*
'That Ramang there, I saw.'

b. { maʔ do ramaŋ **oʔ** }, { wita i ___ }.
mua? do ramaŋ o u-ita i
as for that NAME there 1ERG-see 3ABS *pro*

‘As for that Ramang there, I saw him.’

c. {ι nete kapala ɹesa }, {ι do ramaŋ **oʔ** }, {ι ton djoloʔ ___ }.
na-ita i kapala desa do ramaŋ o tauŋ diolo
3ERG-see 3ABS head village that NAME there year last

‘She saw the mayor, that Ramang there, last year ’

d. { maʔ pole ɹo ramaŋ **oʔ** }, {ι **p**ole toaʔ ___ }.
muaʔ pole i do ramaŋ o pole to aʔ
if come 3ABS that NAME there come also 1ABS

‘If that Ramang there is coming, I’m coming.’

3.4 Phonological Rerouting

To this picture we can add one final fact. Prosodic constituency is routinely reworked in service of output constraints, and at the level of the intonational phrase, we can see this effect in the system of poetry. Mandar has a type of poem called a *Kalinda’da’* that is built around four metrical lines, along the lines in (31).

(31) *The Kalinda’da’*

a. {ι maʔ malo tolaeŋ } {ι pammile malaʔbiʔnaʔ }.
muaʔ maŋ-ala o tolaeŋ pammilei i malaʔbiʔ-na
if ANTIP-take 2ABS someone else choose 3ABS better-3GEN

‘If you’d rather someone else, at least choose someone better.’

b. {ι tane tolaeŋ } {ι maʔ sirrapaŋŋu di }.
tania i tolaeŋ muaʔ siŋ-rapaŋ-u di
is not 3ABS someone else if as-equal-1GEN just.3ABS

‘It’s not someone else if they’re just as bad as me.’ Muthalib & Sangi 1991: 12

To capture the facts of intonational phrasing in these poems, it may seem possible to assume that the metrical lines are separated by some process of the syntax: for instance, the process of clausal extraposition in (33). This is one that operates beneath certain verbs in a lexically restricted way and forces its targets to form intonational phrases.

(32) *Clausal Extraposition*

a. $\{_\iota$ mo ɹo ramam ___ maʔ tallaŋ i manini djoŋiŋ **oʔ** }.
maŋ-ua i do ramaŋ muaʔ tallaŋ i manini dioŋiŋ o
ANTIP-say 3ABS that NAME [$_{CP}$ that sink 3ABS later] yesterday there
'That Ramang there said that it would sink later yesterday.'

b. $\{_\iota$ mo ɹo raman ___ djoŋiŋ **oʔ** } $\{_\iota$ maʔ tallaŋ i manini }.
maŋ-ua i do ramaŋ dioŋiŋ o muaʔ tallaŋ i manini
ANTIP-say 3ABS that NAME yesterday o [$_{CP}$ that sink 3ABS later]
'That Ramang there said yesterday that it would sink later.'

In Mandar, however, it is clear that this analysis is incorrect. The case emerges from a freezing effect: when clauses are extraposed, they become islands for WH-movement. The following example illustrates this fact: WH-movement can escape from complement clauses that remain in the *voice*P (33a) but not the clauses that shift to the right (33b).

(33) *Extraposition: Freezing Effects*

a. $\{_\iota$ a no ɹo rama**t** ___ **t**allam ___ manini djoŋiŋ **oʔ** }?
a na-ua do ramaŋ tallaŋ manini dioŋiŋ o
what 3ERG-say that NAME [$_{CP}$ sink later] yesterday there
'What did that Ramang there say would sink later yesterday?'

b. $\{_\iota$ *a no ɹo raman ___ djoŋiŋ **oʔ** } $\{_\iota$ tallam ___ maniniʔ }?
a na-ua do ramaŋ dioŋiŋ o tallaŋ manini
what 3ERG-say that NAME yesterday there [$_{CP}$ sink later]
INTENDED: 'What did that Ramang there say yesterday would sink later?'

The metrical lines that form their own intonational phrases do not show this effect. As a result, they are transparent to WH-movement: in the following poem, for instance, the WH-phrase *inna* 'where' moves from the second line into the first. I take this fact to suggests that the patterns of intonational phrasing in these poems are forced by purely phonological constraints that govern the number of syllables that appear in each line.

(34) *Wh-Movement: Crosses Metrical Lines*

a. {$_{\iota}$ **inna** wandamo ɹisaŋa } {$_{\iota}$ diposara watammu? ___ }?
inna bandamo di-saŋa di-posara bataŋ-mu
where exactly PASS-think PASS-spoiled body-2GEN

'Where exactly do you think that you will be spoiled?'

b. {$_{\iota}$ atena woŋi } {$_{\iota}$ mala di pattindoaŋ }.
ate-na boŋi mala di pattindoaŋ
heart-3GEN night can in bed

'In the middle of night, it could happen in bed.' Muthalib & Sangi 1991: 284

Reinforcer Postposing reacts to these constraints in a surface-transparent way: when the reinforcers are generated in strings that are parsed into multiple metrical lines, they shift to the edges of their associated lines instead of the positions that they would take without these splits. When the clause in (35a) is split by a linebreak, then, its reinforcer is repositioned in the poem recorded by Muthalib & Sangi 1991 in (35b).

(35) *Phonological Transparency*

a. {$_{\iota}$ jamo **ɹo** ___ ɹisaŋa lopi pattonda roppoŋ **o?** }
jamo do disaŋa lopi pattonda roppoŋ **o** }
is that called boat shipping grass there

'That there is called a worthless endeavor.'

b. {$_{\iota}$ jamo **ɹo** ___ ɹisaŋa **o?** } {$_{\iota}$ lopi pattonda roppoŋ }
jamo do disaŋa **o** } {$_{\iota}$ lopi pattonda roppoŋ }
is that called there boat shipping grass

'{$_{\iota}$ That there is called } {$_{\iota}$ a worthless endeavor }.' Muthalib & Sangi 1991: 374

We can leverage this effect to show that Reinforcer Postposing targets the right edge of the intonational phrase without regard to the syntactic shape of the landing site. By inserting reinforcers into other poems in Muthalib & Sangi 1991, for instance, we can force them to split strings of arguments in the *voice*P (36a), complex DPs (36b), comparatives (36c), and the complements of consider- (36d) and perception-verbs (36e).

(36) *Phonological Rerouting*

a. {$_\iota$ sambo ɹe poppoʔ ___ nawuŋ **eʔ** } {$_\iota$ passon di kappummuʔ }
samboi i de pioppoʔ nauŋ e passauaŋ di kappuŋ-mu
set 3ABS this cover down here well in village-2GEN

'{$_\iota$ Set this cover here down } {$_\iota$ on the well in your village }.

b. {$_\iota$ kaluppiŋa ãʔ **de** salla^{m}bar **eʔ** } {$_\iota$ garritin di lindomuʔ }
kaluppiŋaŋ aʔ de saŋ-lambar e garritiŋ di lindo-mu
wrap up for 1ABS this one-lock here hair from head-2GEN

'{$_\iota$ Wrap up for me this lock here } {$_\iota$ of hair from your head }'

c. {$_\iota$ moŋemoŋe paʔ jau **eʔ** } {$_\iota$ na to nande ɣajaŋ }
moŋe-moŋeʔ pa aʔ iau e na tau na-ande gad͡ʒaŋ
more woe to yet 1ABS me here than person 3ERG-eat sword

'{$_\iota$ Even more woe to [$_{\text{DP}}$ me here] } {$_\iota$ than one mauled by a sword }.

d. {$_\iota$ naparrapam maʔ jau **eʔ** } {$_\iota$ beluwelun di nawaŋ }
na-parrapaŋ mo aʔ iau e belu-beluŋ di nabaŋ
3ERG-consider already 1ABS me here firefly in sky

'{$_\iota$ She already considers [$_{\text{DP}}$ me here] } {$_\iota$ like a firefly in the sky }.

e. {$_\iota$ wita wandi **de** saramu **eʔ** } {$_\iota$ sambar lo ɹi tau }
u-ita bandi de sara-mu e sambar lao di tau
1ERG-see really.3ABS this love-2GEN here slant toward in person

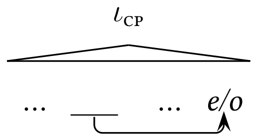
'{$_\iota$ I see this love of yours here } {$_\iota$ falling away toward another }

4 Phonological Movement

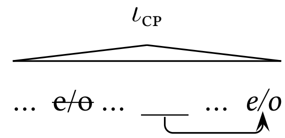
To capture these facts, I propose the analysis in (37): Reinforcer Postposing draws the reinforcers to the edge of the intonational phrase. I take this process to operate in the global, parallel, and constraint-based phonological calculus where prosodic structure is built, following the standard theory of prosodic structure-building (Selkirk 2009, 2011; for overviews, see also Inkelas & Zec 1995; Truckenbrodt 2007; Bennett & Elfner 2019). I propose that Reinforcer Deletion occurs at that stage as well: when the phonology cannot reposition the reinforcers in this way, it prevents them from being exponed.

(37) *The Phonological Analysis*

a. *Reinforcer Postposing*



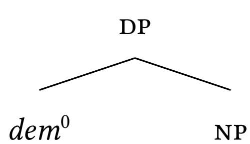
b. *Reinforcer Deletion*



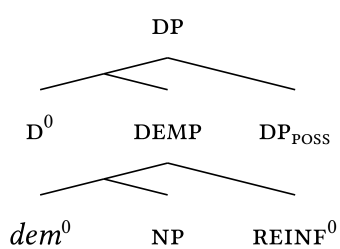
To integrate this result with the theory of linearization, I submit the following proposal. Following the standard view in Distributed Morphology, I assume that linear order is absent from the syntax (Chomsky, 1995) and established in the morphology (Embick, 2007; Kramer, 2010; Myler, 2013). Assuming the ordered morphological module of Arregi & Nevins 2012, I take this process to occur at an intermediate stage of the morphological derivation and establishes ordering relations in response to pressures of three types. The first are the pressures that determine head-complement order, which force heads to precede their complements in Mandar (38a). The second are the pressures that linearize adjuncts and specifiers, which typically force them to fall to the left of their hosts (on which see López 2009b; Kusmer 2019, 2020). The third are the idiosyncratic preferences of particular heads, which have the language-internal effect in (38b): the specifiers that host possessors and reinforcers always surface on the right.

(38) *Morphological Linearization*

a.



b.



I propose that this initial linearization of terminal nodes is reinstantiated over their output constituents in the global and parallel phonological evaluation where prosodic structure is built. Following McCarthy & Prince 1993 and Halpern 1995, I take this process of translation to be driven by ranked and violable constraints. This framework makes it possible for the phonology to directly rework the facts of linear order in response to output constraints: a position that allows us to capture the many patterns of phonological displacement that seem to operate in the prosodic word (Noyer, 1994; Hargus & Tuttle, 1997; Yu, 2007), across the phonological phrase (Halpern, 1992, 1995; Schütze, 1994a,b; King, 1996; Inkelas & Zec, 1990; Zec, 2005; Diesing & Zec, 2011, 2017; Kusmer, 2019), and in larger phonological domains (Bennett *et al.*, 2016; Aissen, 2017).[5] In the system of Reinforcer Postposing, it also allows us to force the step in (39): one that relinearizes the reinforcers to the end of the intonational phrase whenever they are initially linearized into positions that would not fall at that edge.

(39) *Reinforcer Postposing: Phonological Relinearization*

a. *Input Linearization*

ι_{CP}

… *e/o* … ___

b. *Output Linearization*

ι_{CP}

… ___ … *e/o*

On this analysis, we can understand Reinforcer Postposing to reflect a mismatch between the linear order of terminal nodes in the morphology and the linear order of their exponents in the phonology. In the domain of constituency, syntax-prosody mismatches of this this type are routinely forced by output constraints: in the terminology Optimality Theory (Prince & Smolensky, 2004), the restrictions on accentual clash (Kubozono, 1989, 1993; Selkirk & Elordieta, 2010), focus alignment (Féry, 2013, 2017; Féry & Ishihara, 2016), and balance and weight (Hayes & Lahiri, 1991; Ghini, 1993; Shih, 2016; Itô & Mester, 2019). The task of this section is to show that Reinforcer Postposing and Reinforcer Suppression are driven by two output constraints of this type: top-down requirements that are keyed to the right edges of large prosodic domains.

4.1 The Moraic Trochee

The work begins from the system of footing. At the level of the prosodic word, Mandar shows a quantity-sensitive system of foot structure: when top-down constraints do not interfere, the language builds moraic trochees iteratively from the right edge of each prosodic word. These types of feet are built around strings of syllables of four types: cvc, cv-cv, cvc-cv, and cv-cvc (exclusively at the right edge of the prosodic word).[6]

(40) *Quantity-Sensitive Footing*

a. ω	b. ω	c. ω	d. ω
(lèp)(páŋ)	(póle)	(lámba)	(léleŋ)
'stop by'	'come'	'go'	'go about'

These trochees can be seen in many other places in the language: for instance, they are recruited by reduplicative affixes that select adjectives, verbs, and nouns (41a)-(41c).

(41) *The Moraic Trochee*

a.

(**màʔ**)-(màm)(mís),	(**màsi**)-(másiŋ),	(t͡ʃàŋɲo)-(t͡ʃáŋɲo),	(màte)-(máte)
(cvc)-mammis	(cvcv)-masiŋ	(cvccv)-t͡ʃaŋɲo	(cvcv)-mate
RED-sweet	RED-salty	RED-stupid	RED-dead

'A little sweet, a little salty, a little stupid, a little dead.'

b.

(**dàʔ**)-(dàr)(ràs),	(**dùɹu**)-(ɹúɹuŋ),	(dùndu)-(ɹúndu),	(dùku)-(ɹúku)
(cvc)-darras	(cvcv)-duduŋ	(cvccv)-dundu	(cvcv)-duku
RED-recite	RED-carry on head	RED-drink	RED-help

'Recite a little, carry back and forth, drink a little, help repeatedly.'

c.

(**dòʔ**)-(dòt)(tòr),	(**dàt͡ʃi**)-(ɹát͡ʃiŋ),	(dàppa)-(ɹáppa),	(dàli)-(ɹáli)
(cvc)-dottor	(cvcv)-dat͡ʃiŋ	(cvccv)-dappa	(cvcv)-dali
RED-doctor	RED-scale	RED-fathom	RED-earring

'Doctors, toy scales, a few fathoms, various kinds of earrings.'

4.2 The Phonological Phrase

Above the prosodic word and beneath the intonational phrase, there is a single type of intermediate phonological domain that I will term the phonological phrase (Itô & Mester 2007, 2012, 2013; Elfner 2012; *cf.* Nespor & Vogel 1986; Hayes 1989). In Mandar, its right edge is marked by a high tone that falls after low-level syntactic XPs like noun-adjective and verb-adverb strings (42a)-(42b). In regular clauses, it also falls after arguments, phrasal adjuncts, the verbal complex, and the auxiliaries that host agreement (42c).

(42) *The Phonological Phrase*

a.

$\{_{\phi}$	bojak	kes	sannal $\}^{\text{H}}$
	bojaŋ	kaiaŋ	sanna^{l}
	house	big	very

‘A very big house’

b.

$\{_{\phi}$	bemmeʔ	towandi $\}^{\text{H}}$
	bemmeʔ	towandi
	fall	actually

‘Actually fell’

c.

pure$^{\text{H}}$		sita$^{\text{H}}$	ɹotor$^{\text{H}}$	dio$^{\text{H}}$
pura	i	sita	dottor	dio
PAST	3ABS	meet	doctor	there

‘The doctors met there.’

Phonological phrases are also built around many functional projections along the clausal spine in Mandar (Brodkin, 2025a), yielding patterns of recursive phonological phrasing that mirror the syntax below (Selkirk 2009, 2011; Elfner 2012, 2015; Elordieta 2015; Van Handel 2021; see also Ladd 1986; Wagner 2005, 2010; Itô & Mester 2007, 2012, 2013; Ito & Mester 2021). Phrases of this type can be detected from a range of restrictions on segmental processes that emerge in positions medial in the intonational phrase. For instance, there is a process of Voiced Obstruent Lenition that reduces intervocalic */b d d͡ʒ g/* to *[w ɹ j ɣ]*. In three-word strings of the order VSO, this process applies at the left edges of the S and the O 43a). But in four-word VSO strings where the O contains two prosodic words, it is blocked at the left edge of the O (43b). In the same vein, it is always blocked at the left edges of final adjuncts to the *voice*P (43a)-(43b).

(43) *Voiced Obstruent Lenition*

a. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

site^{H}	ɣuru^{H}	ɹotor^{H}	**d**io^{H}
sita i	guru	dottor	dio
meet 3ABS	teacher	doctor	there

'The teacher met the doctor there.'

b. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

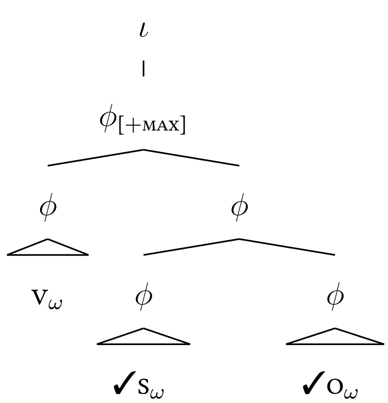
site^{H}	ɣuru^{H}	**d**ottor maliŋgao^{H}	**d**io^{H}
sita i	guru	dottor maliŋgao	dio
meet 3ABS	teacher	doctor tall	there

'The teacher met the tall doctor there.'

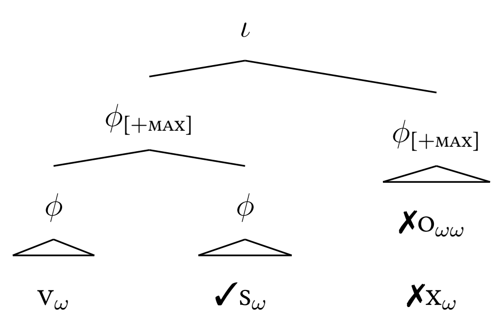
Brodkin 2025a argues that this restriction is keyed to the edge of the type of phonological phrase that is directly dominated by the intonational phrase: the *maximal* phrase phrase, in the terminology of Itô & Mester 2007, 2012, 2013. This analysis is sketched in (44): Voiced Obstruent Lenition proceeds between phonological phrases that are parsed into the same maximal phonological phrase but cannot cross over its edge. We will return to the distribution of the maximal phonological phrase in Section 4.6.

(44) *The Maximal Phonological Phrase*

a. *Left-Edge Lenition*



b. *Prosodic Blocking*



4.3 The Positional Branching Constraint

In phrase-medial positions, the language allows prosodic words to end in monosyllabic feet. This parse emerges in words of the shape cvc (45a)-(45b) and cvccvc (45c)-(45d). In the same positions, words of the shapes cvcv, cvccv, and cvcvc host final disyllabic feet.

(45) *Phrase-Medial Metrical Structure*

a. **(t͡ʃáʔ)** ma(méa)ᴴ
t͡ʃaʔ mamea
paint red
'Red paint'

b. **(léŋ)** (áju)ᴴ
leŋ ad͡ʒu
glue wood
'Wood glue'

c. **(dòt)(tór)** ma(lìŋ)(gáo)ᴴ
dottor maliŋgao
doctor tall
'Tall doctor'

d. **ma(ɣàr)(rít)** to(wándi)ᴴ
magarriŋ tobandi
feverish actually
'Actually feverish'

e. **(gúru)** ma(lìŋ)(gáo)ᴴ
guru maliŋgao
teacher tall
'Tall teacher'

f. **(móŋeʔ)** to(wándi)ᴴ
moŋeʔ tobandi
sick actually
'Actually sick'

At the right edge of the phonological phrase, monosyllabic feet are ruled out. We can see this effect in the phonological phrases that are medial in the intonational phrase but final in the maximal phonological phrase, like those built around arguments and predicates that fall before adjuncts to the *voice*ᴘ. At the edges of those phrases, several repairs conspire to build final disyllabic feet. Cvc roots undergo ʔv-epenthesis, which copies their vowels, infixes them into the word, and breaks hiatus with a glottal stop.[7]

(46) *ʔv-Epenthesis*

a. {$_{\phi[\text{MAX}]}$ nalaŋaᴴ ãʔ **(t͡ʃáʔaʔ)**ᴴ } {$_{\phi[\text{MAX}]}$ díoᴴ. }
na-alaŋaŋ aʔ t͡ʃaʔ dio
3ᴇʀɢ-get for 1ᴀʙs paint there
'They got me some paint there.'

b. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

nalaŋa$^{\text{H}}$ ãʔ **(léʔen)**$^{\text{H}}$ diɣénaʔ$^{\text{H}}$.
na-alaŋaŋ aʔ leŋ digenaʔ
3ERG-get for 1ABS glue earlier
'She got me some glue earlier.'

Final-cvccvc roots are forced to host the same types of feet, and to this end they undergo five analogous repairs. Final glottal stops delete in roots of the shape cvccvʔ (47a), medial codas delete in roots that host medial geminates, nasals, and glottal stops (47b)-(47d), and the final lateral deletes in the single root *sannal* 'very' (47e).

(47) *Segmental Deletions*

a. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

póle$^{\text{H}}$ **i(t͡ʃítt͡ʃi)**$^{\text{H}}$ djóŋiŋ$^{\text{H}}$.
pole i it͡ʃitt͡ʃiʔ dioŋiŋ
come 3ABS NAME yesterday
'Chichi' came yesterday.'

b. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

néte$^{\text{H}}$ **(ɹótor)**$^{\text{H}}$ diràmbóŋi$^{\text{H}}$.
na-ita i dottor diruamboŋi
3ERG-see 3ABS doctor a while ago
'The doctor saw him a while ago.'

c. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

píto$^{\text{H}}$ **(pá$^{\text{n}}$des)**$^{\text{H}}$ sàllíwaŋ$^{\text{H}}$.
peita o pandeŋ sallibaŋ
go look for 2ABS pineapple on the other side
IDIOMATICALLY: 'Go look for a wife in another village.'

d. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

píto$^{\text{H}}$ **(kúmil)**$^{\text{H}}$ sàllíwaŋ$^{\text{H}}$.
peita o kuʔmil sallibaŋ
go look for 2ABS jackfruit on the other side
'Go look for jackfruit outside.'

e. {$_{\phi[\textsc{max}]}$ } {$_{\phi[\textsc{max}]}$ }

bjase$^{\text{H}}$		matʃ͡oa	**(sánna)**$^{\text{H}}$	dío$^{\text{H}}$.
bjasa	i	matʃ͡oa	sanna^{l}	dio
usually	3ABS	good	very	there

‘It’s usually very good there.’

These effects reveal that the phonology requires the final foot in the phonological phrase to be the disyllabic trochee in (48). This is a top-down requirement for binarity (Booij 1999; Prieto 2005; Elordieta 2006), first documented in another corner of the language by Brodkin 2025b, and I will refer to it as the Positional Branching Constraint.

(48) *The Positional Branching Constraint*

a. ϕ

… ✓($\acute{\sigma}\sigma$)

b. ϕ

… ✗($\acute{\sigma}$)

4.4 Positional Licensing

At the right edge of the intonational phrase, we see a second effect: the language suspends every repair to the Positional Branching Constraint. At the edges of phrases in that position,cv(c) roots remain monosyllabic and cvccvc-final roots show no deletion.[8]

(49) *The Suspension Effect*

a. {$_{\iota}$

nalájna$^{\text{H}}$	ãʔ	**(léɲ)**$^{\text{H}}$ }.
na-alaɲaɲ	aʔ	leɲ
3ERG-get for	1ABS	glue

‘She got me some glue.’

b. {$_{\iota}$

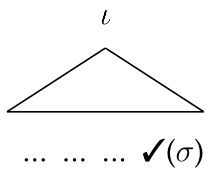
néte$^{\text{H}}$		**i(tʃ͡ìt)(tʃ͡íʔ)**$^{\text{H}}$ }.
na-ita	i	itʃ͡ittʃ͡iʔ
3ERG-see	3ABS	NAME

‘Chichi’ saw him.’

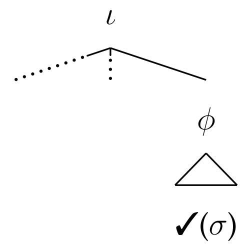
These effects trace a second top-down constraint: the intonational phrase preferentially hosts a final monosyllabic foot (50a) and supersedes the Positional Branching Constraint to license phrase-final monosyllabic feet in this position (50b).

(50) *Positional Licensing*

a. *The Final Foot*



b. *Suspending Minimality*



The case for this constraint emerges from a process of ʔ-epenthesis that operates at the right edge of the intonational phrase. Outside the right edge of that domain, words that contain odd numbers of light syllables typically receive the non-exhaustive parse in (51a). When those words surface at the right edge of the intonational phrase, however, they undergo a process of ʔ-epenthesis and receive the exhaustive parse in (51b). The same process never applies to roots that contain even numbers of light syllables, which are exhaustively parsed in the default case (51c)-(51d).

(51) *Glottal Epenthesis*

a. {$_ι$ démmo$^{\text{H}}$ **la(súna)**$^{\text{H}}$ áʔ$^{\text{H}}$ }.
diaŋ mo lasuna a
there is now shallot PRT
'Now there are shallots, eh?'

b. {$_ι$ démmo$^{\text{H}}$ **(làsu)(náʔ)**$^{\text{H}}$ }.
diaŋ mo lasuna
there is now shallot
'Now there are shallots.'

c. {$_ι$ démmo$^{\text{H}}$ **(lémo)**$^{\text{H}}$ áʔ$^{\text{H}}$ }.
diaŋ mo lemo a
there is now citrus PRT
'Now there are lemons, eh?'

d. {$_ι$ démmo$^{\text{H}}$ **(lémo)**$^{\text{H}}$ }.
diaŋ mo lemo
there is now citrus
'Now there are lemons.'

In words of different shapes, this process applies whenever it can create a monosyllabic foot without derailing exhaustive footing through the rest of the word. We can see this effect in the behavior of three types of words that typically receive an exhaustive parse: those of the shapes cvc-cv, cvc-cv-cv, and cvc-cvc-cv. Outside of this edge, these words typically host final disyllabic feet—but at the right edge of the intonational phrase, they undergo ʔ-epenthesis and host final monosyllabic feet (52).

(52) *Final Monosyllabic Feet*

a. {$_{\iota}$ démmo$^{\text{H}}$ **(ánde)**$^{\text{H}}$ áʔ$^{\text{H}}$ }.

diaŋ	mo	ande	a
there is	now	food	PRT

'Now there's food, eh?'

b. {$_{\iota}$ démmo$^{\text{H}}$ **(àn)(déʔ)**$^{\text{H}}$ }.

diaŋ	mo	ande
there is	now	food

'Now there's food.'

c. {$_{\iota}$ démmo$^{\text{H}}$ **(àn)(d͡ʒóro)**$^{\text{H}}$ áʔ$^{\text{H}}$ }.

diaŋ	mo	and͡ʒoro	a
there is	now	coconut	PRT

'Now there are coconuts, eh?'

d. {$_{\iota}$ démmo$^{\text{H}}$ **(ànd͡ʒo)(róʔ)**$^{\text{H}}$ }.

diaŋ	mo	and͡ʒoro
there is	now	coconut

'Now there are coconuts.'

e. {$_{\iota}$ démmo$^{\text{H}}$ **(àk)(kátta)**$^{\text{H}}$ áʔ$^{\text{H}}$ }.

diaŋ	mo	akkatta	a
there is	now	plan	PRT

'Now there's a plan, eh?'

f. {$_{\iota}$ démmo$^{\text{H}}$ **(àk)(kàt)(táʔ)**$^{\text{H}}$ }.

diaŋ	mo	akkatta
there is	now	plan

'Now there's a plan.'

4.5 Reinforcers Revised

This footwork reveals that Reinforcer Postposing and Reinforcer Deletion have an optimizing effect. The reinforcers form words and phrases of their own: like possessors in SPEC,DP and phrasal adjuncts to the DP, they host word-level stress and carry phrase-final high tones. As a result, they implicitly violate the Positional Branching Constraint: the reinforcers form phonological phrases and cannot host disyllabic feet.

(53) *The Reinforcers: Phrase-Final Monosyllabic Feet*

a.

do	wúku$^{\text{H}}$	dío$^{\text{H}}$	**(óʔ)**$^{\text{H}}$
do	buku	dio	o
that	book	there	there

'That book there'

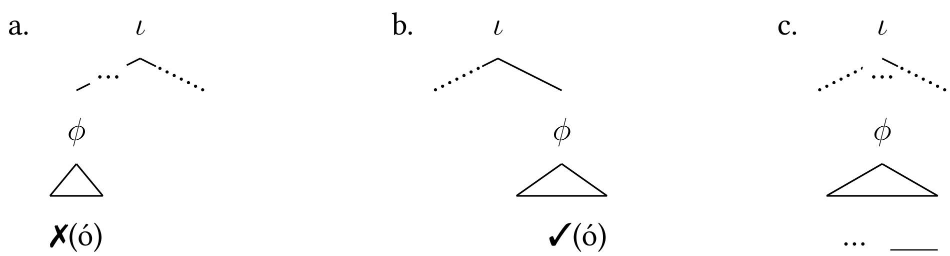
b.

de	wukúna$^{\text{H}}$	ikát͡ʃoʔ$^{\text{H}}$	**(éʔ)**$^{\text{H}}$
de	buku-na	ikat͡ʃoʔ	e
this	book-3GEN	NAME	here

'This book here of Kacho's.'

As a result, we can understand Reinforcer Postposing and Reinforcer Deletion as output-oriented repairs to the system of Positional Binarity: Reinforcer Postposing draws the reinforcers to positions that license phrase-final monosyllabic feet (54b) and Reinforcer Deletion suppresses them whenever Postposing cannot apply (54c).

(54) *Reinforcers and Positional Minimality*



4.6 Syllable Attraction

The central argument for this analysis emerges from a suspension effect: Reinforcer Postposing and Reinforcer Deletion alternate transparently with a second phonological repair. This is one that emerges in positions that are medial in the maximal phonological phrase. In this environment, the Positional Branching Constraint is repaired in two different ways depending on the metrical status of the syllable at the left edge of the following phonological phrase: we see different repairs before positions where that syllable should be footed, like (55a), and positions where it should not, like (55b).

(55) *The Positional Divide*

In the first context, we see our usual repairs: before roots that are disyllabic (56a) and roots that begin with initial heavy syllables (56b), final-cvccvc roots show the usual patterns of reduction and cvc roots show the usual process of ʔv-epenthesis.

(56) *Regular Repairs in the Maximal Phonological Phrase*

a. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

néte$^{\text{H}}$		**(ɹótor)**$^{\text{H}}$	**(gúru)**$^{\text{H}}$	dío$^{\text{H}}$.
na-ita	i	dottor	guru	dio
3ERG-see	3ABS	doctor	teacher	there

'The doctor saw the teacher there.'

b. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

wéŋa$^{\text{H}}$	ĩ	**(léʔek)**$^{\text{H}}$	**(kàn)(díʔu)**$^{\text{H}}$	diɣénaʔ$^{\text{H}}$.
u-beŋaŋ	i	leŋ	kandiʔ-u	digenaʔ
1ERG-give	3ABS	glue	sibling-1GEN	earlier

'I gave some glue to my little sibling earlier.'

In the second position, we see two shifts. First, before trisyllabic roots that begin in light syllables, the phrase-final high tone shifts one syllable to the right. Second, the usual repairs cease: cvc roots show no epenthesis (57a), cvccvc roots retain final glottal stops and all types of medial codas (57b)-(57d), and *sannal* retains its lateral (57e).

(57) *The Special Repair*

a. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

wéŋa$^{\text{H}}$	ĩ	**lék**	**ka**$^{\text{H}}$(káʔu)$^{\text{H}}$	diɣénaʔ$^{\text{H}}$.
u-beŋaŋ	i	leŋ	kakaʔ-u	digenaʔ
1ERG-give	3ABS	glue	sibling-1GEN	earlier

'I gave some glue to my older sibling earlier.'

b. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

néte$^{\text{H}}$		**i(t͡ʃit)t͡ʃíʔ**	**gu**$^{\text{H}}$(rúna)$^{\text{H}}$	dío$^{\text{H}}$.
na-ita	i	it͡ʃitt͡ʃiʔ	guru-na	dio
3ERG-see	3ABS	NAME	teacher-3GEN	there

'Chichi' saw her teacher there.'

c. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

néte$^{\text{H}}$		**(ɹòt)tór**	**gu**$^{\text{H}}$(rúʔu)$^{\text{H}}$	dío$^{\text{H}}$.
na-ita	i	dottor	guru-u	dio
3ERG-see	3ABS	doctor	teacher-1GEN	there

'The doctor saw my teacher there.'

d. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

waláŋa$^{\text{H}}$	ĩ	**(pàn)dék**	**ka**$^{\text{H}}$(káʔu)$^{\text{H}}$	dío$^{\text{H}}$.
u-alaŋaŋ	i	pandeŋ	kakaʔ-u	dio
1ERG-get for	3ABS	pineapple	sibling-1GEN	there

'I got some pineapple for my older sibling there.'

e. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

waláŋa$^{\text{H}}$	ĩ	**(kùʔ)míl**	**ka**$^{\text{H}}$(káʔu)$^{\text{H}}$	dío$^{\text{H}}$.
u-alaŋaŋ	i	kuʔmil	kakaʔ-u	dio
1ERG-get for	3ABS	jackfruit	sibling-1GEN	there

'I got some jackfruit for my older sibling there.'

f. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

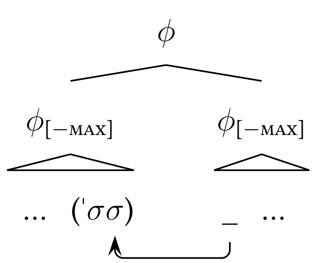
nd́e		pa$^{\text{H}}$lákam$^{\text{H}}$	matʃ͡óa	**(sàn)nál**	**d͡ʒe**$^{\text{H}}$(páʔna)$^{\text{H}}$	dítiŋ$^{\text{H}}$.
ndaŋ	i	palakaŋ	matʃ͡oa	sanna$^{\text{l}}$	d͡ʒepaʔ-na	ditiŋ
not	3ABS	turns out	good	very	tortilla-3GEN	there

'Turns out the tortillas weren't very good there.'

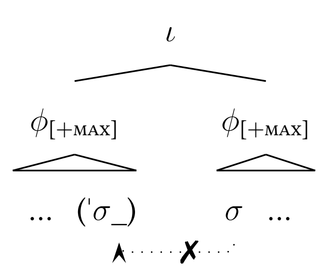
These effects reveal that the phonology preferentially constructs this disyllabic foot around roots of this shape by drawing in syllables that would otherwise go unfooted from the left edge of the following phonological phrase (58a). This is one of many processes in the phonology of the language that is blocked at the right edge of the maximal phonological phrase (58b), and its wider distribution has been documented by Brodkin 2025b. I will refer to it as SYLLABLE ATTRACTION below.

(58) *Syllable Attraction*

a. *The Optimal Repair*

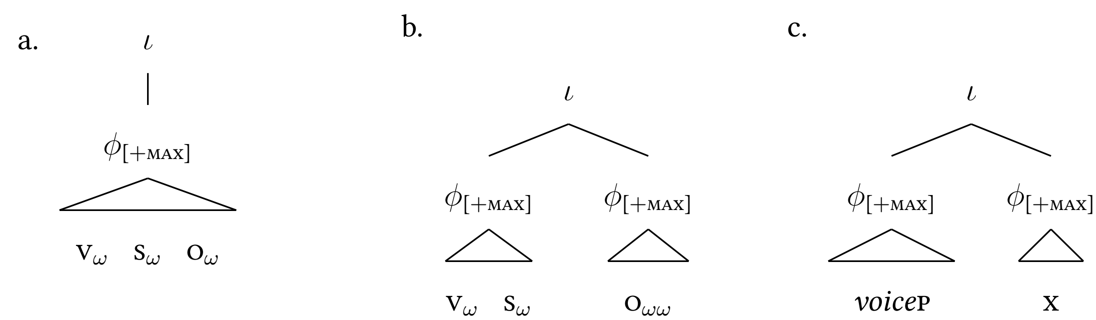


b. *The Top-Down Constraint*



To see how the reinforcers interact with this repair, we can return to the distribution of the maximal phonological phrase. In Mandar, maximal phonological phrases are typically built around strings of verbs and arguments that contain two to three prosodic words. The following diagrams summarize this effect: maximal phonological phrase are built around three-word strings of the orders VSO, VSD, and VOD (where D = dative argument; (59a)), they are split into multiple maximal phonological phrases when they contain more than three words (59b), and every adjunct to the *voice*P is parsed into a maximal phonological phrase of its own (59c) (Brodkin 2025a; see also Hale & Selkirk 1987; Chen 1987; Truckenbrodt 1995, 1999).

(59) *Maximal Phonological Phrasing*



From the facts of Syllable Attraction, we can see how maximal phonological phrases are realigned when we introduce additional prosodic words into arguments that are medial in the *voice*P. In VOD strings where the O contains two words, it cannot force Syllable Attraction into the following D (60a)—suggesting that the two are not parsed into the same maximal phonological phrase. When we introduce additional prosodic words to the verbal complex in this context, however, the O begins to trigger Syllable Attraction (60b). This is one of many effects that reveals the following generalization: when the verb contains two prosodic words, two-word arguments that follow are parsed into maximal phonological phrases with one-word arguments to their right.

(60) *The Weight Constraint*

a. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

wéŋa^{H}	ĩ	wúku	**(kétt͡ʃu)**^{H}	**gu**(rúʔu)^{H}	diɣénaʔ^{H}.
u-beŋaŋ	i	buku	kett͡ʃuʔ	guru-u	digenaʔ
1ERG-give	3ABS	book	little	teacher-1GEN	earlier

'I gave a little book to my teacher earlier.'

b. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

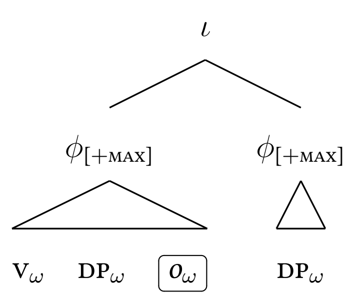
mé	wéŋaŋ^{H}	i	búku	ket(**t͡ʃúʔ**	**gu)**^{H}(rúʔu)^{H}	diɣénaʔ^{H}.
mane	u-beŋaŋ	i	buku	kett͡ʃuʔ	guru-u	digenaʔ
just	1ERG-give	3ABS	book	little	teacher-1GEN	earlier

'I gave a little book to my teacher earlier.'

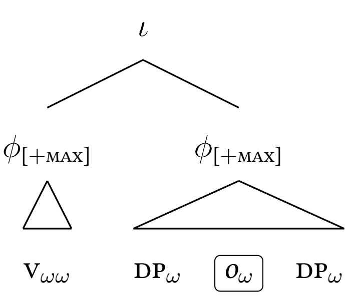
This footwork sets up the following prediction. As the reinforcers form prosodic words, they should trigger the same prosodic shift as the adjectives above. When a reinforcer is associated with a one-word argument that is medial in the *voice*P, more specifically, we should see the following alternation. When the verbal complex contains a single word, the phonology should never build maximal phonological phrases around four-word strings like V-O-REINFORCER-D (61a). But when the verbal complex contains two words, the phonology should build maximal phonological phrases around strings like O-REINFORCER-D (61b). In that second context, Syllable Attraction should occur.

(61) *The Weight Constraint*

a. *Two-Word Arguments*



b. *The Manipulation*



This prediction is correct. When the reinforcers are initially linearized into positions that would force them to fall at the right edges of maximal phonological phrases if they remained in-situ, they move or delete (62a). When they are linearized into positions that should allow Syllable Attraction, however, they surface in place (62b).

(62) *Reinforcers Trigger Syllable Attraction*

a. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

wéŋa^{H}	ĩ	ɹo	wúku^{H}	__	ɣurúʔu^{H}	diɣénaʔ^{H}	óʔ^{H}.
u-beŋaŋ	i	do	buku		guru-u	digenaʔ	o
1ERG-give	3ABS	that	book		teacher-1GEN	earlier	there

'I gave that book there to my teacher earlier.'

b. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

mé	wéŋaŋ^{H}	i	do	wúku^{H}	(**ó**	**ɣu)**^{H}(rúʔu)^{H}	diɣénaʔ^{H}.
mane	u-beŋaŋ	i	do	buku	o	guru-u	digenaʔ
just	1ERG-give	3ABS	that	book	there	teacher-1GEN	earlier

'I just gave that book there to my teacher earlier.'

With the same manipulations, we can force the reinforcers to follow their associates in any position that allows them to force Syllable Attraction. In complex DPs, for instance, we can force them to surface immediately after their associates by introducing rightward specifiers that contain exactly one word and begin with unstressed syllables (63a). In the left periphery, in turn, we can force them to follow associates in the same positions that host WH-phrases and allow Syllable Attraction into the verb (63b).

(63) *Reinforcers in-Situ*

a. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

do	wukúna^{H}	(**ó**	**ɣu)**^{H}(rúʔu)^{H}	dó	ɹi	méja^{H}.
do	buku-na	o	guru-u	dio	di	med͡ʒa
that	book-3GEN	there	teacher-1GEN	there	on	table

'That book of my teacher's there on the table.'

b. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

do wúku^{H} (**ó** **na)**^{H}(wémme)^{H} manini^{H}.
do buku o na-bemmeʔ manini
that book there will-fall later

'That book there will fall later.'

When we rule out Syllable Attraction in this second configurations, the reinforcers lose the ability to remain in-situ. In VOD strings where the object bears initial stress, the reinforcers that are generated in the S are forced to move (64a) or delete (64b). In the complex DPs where their initial positions are followed by specifiers that begin with stressed syllables or host two prosodic words, we see the same effects (64c)-(64d).

(64) *Phonological Disruptions*

a. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

máne wéŋaŋ^{H} i do wúku^{H} __ **(ɣúru)**^{H} diɣénaʔ^{H} óʔ^{H}.
mane u-beŋaŋ i do buku guru-u digenaʔ o
just 1ERG-give 3ABS that book teacher earlier there

'I just gave that book there to my teacher earlier.'

b. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

máne wéŋaŋ^{H} i do wúku^{H} __ **(ɣúru)**^{H} díni^{H} éʔ^{H}.
mane u-beŋaŋ i do buku o guru-u dini e
just 1ERG-give 3ABS that book there teacher here here

'I just gave that book there to my teacher here.'

c. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

do wukúna^{H} __ **(ɣúru)**^{H} dó ɹi méja^{H} óʔ^{H}.
do buku-na guru dio di med͡ʒa o
that book-3GEN teacher there on table there

'That book of the teacher's there on the table.'

d. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

do wukúna^{H} __ **ɣu(rúna)**^{H} ikát͡ʃoʔ díni ɹi méja^{H} éʔ^{H}.
do buku-na o guru-na ikat͡ʃoʔ dini di med͡ʒa e
that book-3GEN there teacher-3GEN NAME here on table here

'That book of the teacher's here on the table.'

When the reinforcers are able to stay in-situ, they are freely crossed by reinforcers initially linearized further to the left. In this context, in other words, they are phonologically deactiviated and cease to intervene in the system of relative locality.

(65) *Phonological Deactivation*

a. {ϕ[MAX] } {ϕ[MAX] } {ϕ[MAX] }

de wúku kés sànnálna$^{\text{H}}$ __ do solána$^{\text{H}}$ (**ó** **i**)$^{\text{H}}$(kát͡ʃoʔ)$^{\text{H}}$ (**éʔ**)$^{\text{H}}$.
de buku kaiaŋ sanna$^{\text{l}}$-na do sola-na o ikat͡ʃoʔ e
this book big very-3GEN that friend-3GEN there NAME here

'This really big book here of that friend there of Kacho's.'

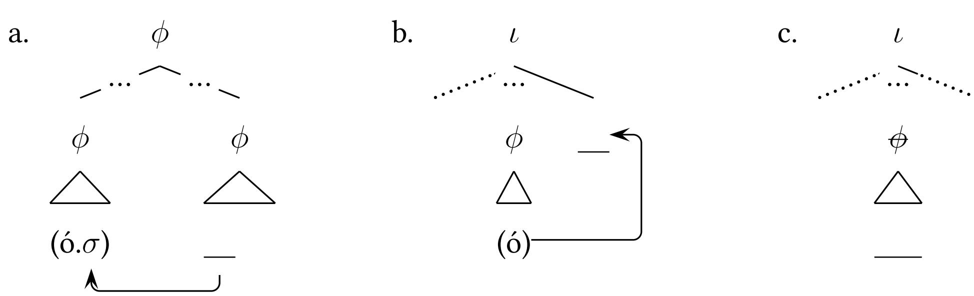
b. {ϕ[MAX] } {ϕ[MAX] } {ϕ[MAX] }

mé nawéŋa$^{\text{H}}$ ĩ ɹe sànaéke$^{\text{H}}$ __ do wúku$^{\text{H}}$ (**ó** **ɣ**)**u**$^{\text{H}}$(rúna)$^{\text{H}}$ (**éʔ**)$^{\text{H}}$.
mane na-beŋaŋ i de sanaeke do buku o guru-na e
just 3ERG-give 3ABS this kid that book there teacher-3GEN here

'This kid here just gave that book there to his teacher.'

The result is a decisive case that Reinforcer Postposing and Reinforcer Deletion form part of the system of Positional Minimality: as they are bled by Syllable Attraction, they must occur in the phonology and be forced by the Positional Branching Constraint. We can thus understand them as a ranked series of repairs to rescue the reinforcers from their internal tension: when possible, they trigger Syllable Attraction (66a), when they cannot, they shift (66b), and when that repair is ruled out, they are not exponed (66c).

(66) *Item-Specific Repairs*



5 Linearization and Output Optimization

To capture the facts of relinearization, I propose that the linear order of the morphology is guarded by the isomorphism constraint in (67), which checks the linear order of pairs of terminal nodes in the input and compares it against the order of their exponents in the output (and see also Halpern 1995; Göbbel 2007; López 2009a; Bennett *et al.* 2016; Aissen 2017; Clemens & Coon 2018; Clemens 2019; Kusmer 2019; Potsdam 2022).

(67) No Shift
Assign one violation for every pair of constituents α, β, such that α is linearized before β in the morphological input, for which the output correspondent of α is not linearized before the output correspondent of β in the phonological output.

5.1 Forcing Movement

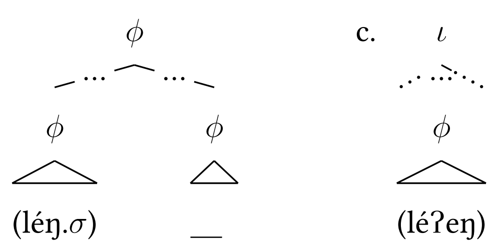
The repairs to the Positional Branching Constraint emerge from the relative ranking of No Shift with other constraints. The behavior of the cvc roots follows from the interaction of NoShift with dep, which prevents epenthesis, and the Align constraints, which force isomorphism between morphosyntactic and prosodic domains (McCarthy & Prince 1993; Selkirk 1995a,b). With the ranking in (68a), we can force them to trigger Syllable Attraction when they can (68b) and show epenthesis when they cannot (68c).

(68) *Syllable Attraction*

a.

No Shift
|
Dep
|
Align

b. c.



To force Reinforcer Postposing, in turn, we can introduce two further constraints. The first is the lexically indexed version of DEP in (69a), which prevents ʔv-epenthesis in the reinforcers (for discussion: Pater 2000, 2006, 2008; Gouskova 2012). The second is the TETHERING constraint in (69b), which prevents the reinforcers from shifting to the left when that would provide a shorter path to the right edge of an intonational phrase.

(69) *Reinforcers: Constraints*

a. DEP-V$_{e/o}$

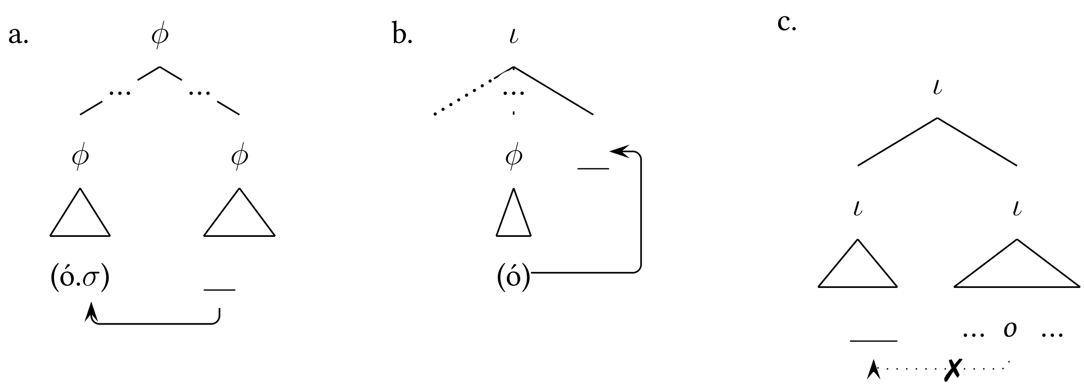
Assign one violation for every vocalic segment in the output that has no correspondent in the input in the exponents of $\sqrt{\text{HERE}}$ and $\sqrt{\text{THERE}}$.

b. TETHERING

Assign one violation for every demonstrative-reinforcer pair α, β, such that α selects β in the morphological input, for which the output correspondents of α and β are not in the same intonational phrase in the phonological output.

By the ranking these constraints above NO SHIFT and ALIGN, we can ensure that the reinforcers will trigger Syllable Attraction whenever they can (70a), move when they cannot (70b), and never escape the intonational phrase (70c).

(70) *Reinforcer Postposing*



5.2 Deriving the Path

To derive the path of Reinforcer Postposing, at last, we can introduce two types of MATCH constraint (and see especially (Elfner, 2012, 2015; Bennett *et al.*, 2016)). The first is the MAX constraint in (71a), which forces the construction of phonological phrases that contain the exponents of all the terminal nodes in their corresponding XPs. The second is the DEP constraint in (71b), which prevents the formation of phonological phrases that contain the exponents of terminal nodes not contained in their corresponding XPs.

(71) *Match Constraints*

a. MATCH-XP

Assign one violation for every XP that dominates a set of terminal nodes in the morphological input that does not correspond to a phonological phrase that contains the output correspondents of all of those terminal nodes in the phonological output.

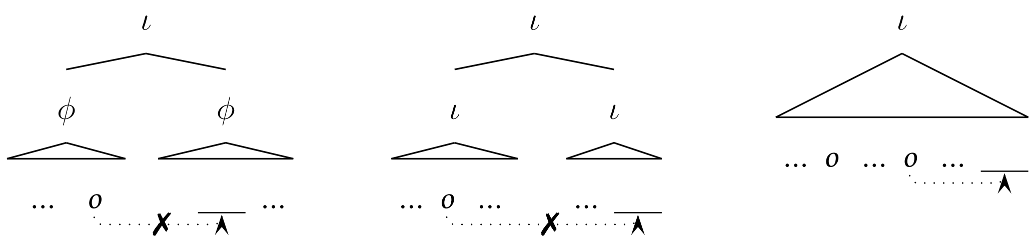
b. MATCH-ϕ

Assign one violation for every phonological phrase in the phonological output that contain the output correspondents of terminal nodes not contained in its corresponding XP in the morphological input.

These constraints allow us to force the reinforcers to shift to the right edge of the intonational phrase and form maximal phonological phrases of their own. By ranking NOSHIFT over the MAX constraint, to begin, we can force the reinforcers to shift when they have no other escape from the Positional Branching Constraint. By ranking DEP over NOSHIFT, in turn, we can prevent them from simply shifting into the linearly closest position that should allow Syllable Attraction (72a). From this position, at last, we can derive two further results from violation profile of NOSHIFT itself: as this constraint imposes gradient violations that are sensitive to linear order, it will always force movement to the edge of the smallest containing intonational phrase (72b) and force movement of the needy reinforcers that are linearized furthest to the right (72c).[9]

(72) *Deriving the Path*

a. *Absolute Locality* b. *Absolute Locality* c. *Relative Locality*



6 Suspended Exponence

To capture the facts of Reinforcer Deletion, I propose that exponence proceeds through a two-step process that is distributed between the morphology and the phonology. Following the standard view in Distributed Morphology, I take morphological exponent selection to occur in a cyclic and bottom-up process that targets terminal nodes in a series of steps that follow impoverishment (Halle & Marantz, 1993), node insertion and deletion, fission and fusion (Hewett, 2023), and linearization (Adger 2006; Merchant & Pavlou 2017; Ostrove 2018; see also Embick & Noyer 2001). I assume that it proceeds in three steps that I will refer to collectively as the operation Match. First, it identifies the complete set of exponents that can be matched with the target head. Second, it narrows down the set of possible exponents on the basis of the subset principle (Halle & Marantz, 1993), which favors the candidates that most closely match the featural specification and position of the target (Bobaljik, 2000, 2012; Embick, 2010; Gouskova & Bobaljik, 2020). Third, when multiple candidates are favored to the same degree, it narrows the set further on the basis of morphological markedness constraints (Arregi & Nevins, 2012).

(73) *Stage One: Match*

a. *Possibilities*

$$x^0 \\ \triangle \\ [\alpha][\beta][\gamma]$$

b. *Subset Principle*

$$x^0 \\ \triangle \\ [\alpha][\beta]$$

c. *Markedness*

$$x^0 \\ \triangle \\ [\alpha]$$

I propose that exponents are inserted into terminal nodes by a second process, INSERT, which occurs in the phonology and proceeds in parallel with the translation of constituent structure and linear order. In the simplest cases, these systems translate linearized and unexponed morphological representations like (74a) to linearized and exponed prosodic representations like (74b). But in contexts of conflict with output constraints, I propose that the phonology can force the type of suspended exponence in (74c): one that directly suspends insertion to satsify output constraints (and see also Raffelsiefen 1999, 2004; Kurisu 2001; Henderson 2012, and Aissen 2017).

(74) *Stage Two: Insert*

a. *Morphology*

$$\text{XP} \\ | \\ x^0 \\ \triangle \\ [\alpha]$$

b. *Phonology*

$$\phi \\ | \\ \omega \\ \triangle \\ [\alpha]$$

c. *Suspension*

$$\cancel{\phi} \\ | \\ \cancel{\omega} \\ \triangle \\ \_\_\_$$

6.1 Beyond the Reinforcers

To scaffold this account, we can observe that Reinforcer Deletion is matched by an analogous repair that targets a verbal suffix: the transitivity marker *-í*. This suffix is exponed in phrase-medial positions (75a), at the right edge of the intonational phrase

(75b), and in the phrase-final positions that allow Syllable Attraction (75c).

(75) *The Transitivity Suffix*

a.

$\{_{\phi}$	wóloʔ**(í)**	towándi	$\}^{\text{H}}$ $\{_{\phi}$	déʔ	$\}^{\text{H}}$.
	u-oloʔ-i	tobandi		de	
	1ERG-like-I	actually.3ABS		I guess	

'I actually like it, I guess.'

b.

$\{_{\iota}$ bése$^{\text{H}}$		palákam$^{\text{H}}$	**wóloʔ(íʔ)**$^{\text{H}}$ }.	
biasa	i	palakaŋ	u-oloʔ-i	
usually	3ABS	turns out	1ERG-like-I	

'Turns out I usually like it.'

c.

$\{_{\phi[\text{MAX}]}$		}	$\{_{\phi[\text{MAX}]}$	}	$\{_{\phi[\text{MAX}]}$ }
bése$^{\text{H}}$		palákam$^{\text{H}}$	wóloʔ**(í**	**je)**$^{\text{H}}$(páʔna)$^{\text{H}}$	dío$^{\text{H}}$.
biasa	i	palakaŋ	u-oloʔ-i	d͡ʒepaʔ-na	dio
usually	3ABS	turns out	1ERG-like-I	tortilla-3GEN	there

'Turns out I usually like their tortillas there.'

In the contexts that prohibit monosyllabic feet and Syllable Attraction, this suffix is not exponed. As a result, it disappears at the edges of maximal phonological phrases (76a) and non-maximal phonological phrases that precede footed syllables (76b).

(76) *Suspended Exponence*

a.

$\{_{\phi[\text{MAX}]}$			}	$\{_{\phi[\text{MAX}]}$ }
bése$^{\text{H}}$		palákam$^{\text{H}}$	**wóloʔ**$^{\text{H}}$	dío$^{\text{H}}$.
biasa	i	palakaŋ	u-oloʔ-i	dio
usually	3ABS	turns out	1ERG-like-I	there

'Turns out I usually like it there.'

b.

$\{_{\phi[\text{MAX}]}$		}	$\{_{\phi[\text{MAX}]}$	}	$\{_{\phi[\text{MAX}]}$ }
bése$^{\text{H}}$		palákam$^{\text{H}}$	**wólo**$^{\text{H}}$	(ánde)$^{\text{H}}$	dío$^{\text{H}}$.
biasa	i	palakaŋ	u-oloʔ-i	ande	dio
usually	3ABS	turns out	1ERG-like-I	food	there

'Turns out I usually like the food there.'

The same effect is forced in phrase-medial positions by the Rhythm Rule, which prevents monosyllabic feet from surfacing before words that bear initial stress. Examples (77a) shows its basic effect with a final-cvccvc root: before the adverb *bandi* 'really', it takes penultimate stress and hosts a final disyllabic foot. Example (77b) then shows how the suffix *i* reacts to this constraint: in this prosodic context, it is suppressed.

(77) *The Rhythm Rule*

a. $\{_{\phi}$ mat͡ʃóa (**sánna**) (wán)(díʔ) $\}^{H}$
mat͡ʃoa sanna^{l} bandi
good very really
'Is it really very good?'

b. $\{_{\phi}$ **móloʔ**____ (wàn)(díʔ) $\}^{H}$
mu-oloʔ-i bandi
2ERG-like-I really
'Do you really like it?'

6.2 The Violable Constraint

To capture these effects, I propose that INSERT is forced by a ranked and violable constraint: the version of MAX in (78), which forces insertion to occur (and see also Raffelsiefen 1996, 1999, 2004; Kurisu 2001; Henderson 2012).

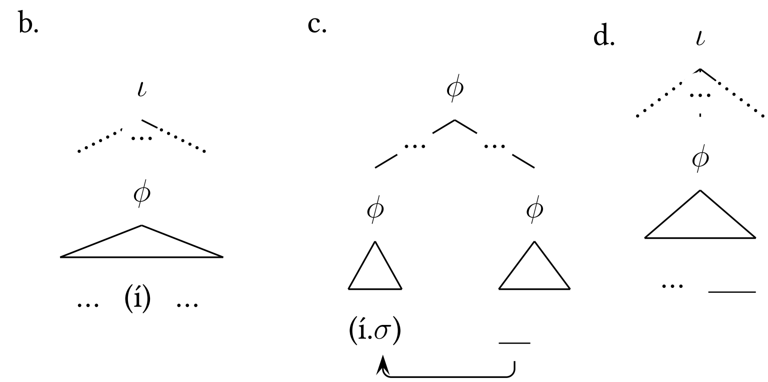
(78) EXPONE

Assign one violation for every terminal node in the input that is not targeted by INSERT to receive a correspondent in the output.

Suspended exponence of the transitivity suffix emerges from the ranking in (79a). When the morphological process of MATCH selects the exponent *i* for this head, the phonology inserts it in the contexts that allow monosyllabic feet (79b). In the contexts that allow Syllable Attraction, the ranking of EXPONE over ALIGN forces the same result (79c). In other contexts, the ranking of the Positional Minimal Constraint and an item-specific DEP constraint over EXPONE and NOSHIFT forces another repair. I assume that movement is ruled out in this context by higher-ranking MATCH constraints that operate at the level of the word and prevent *-i* from being parsed into any other prosodic word. As a result, I take these constraints to force the type of non-exponence in (79d).

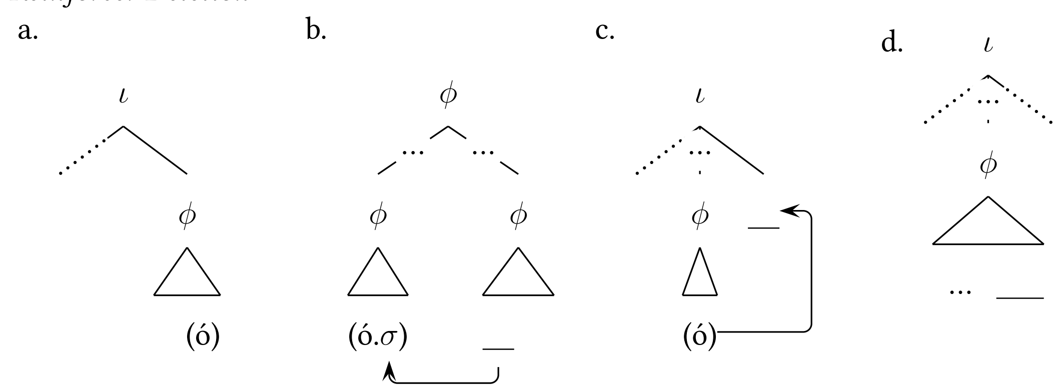
(79) *Suffix Suppression*

a.
MATCH-ω
|
DEP$_i$
|
EXPONE
|
NO SHIFT
|
ALIGN



Reinforcer Deletion emerges in exactly the same way. EXPONE forces the reinforcers to be realized in the contexts that allow monosyllabic feet (80a). Whenever Syllable Attraction is possible, the ranking of EXPONE and NO SHIFT over ALIGN forces that repair (80b). When Syllable Attraction is impossible, in turn, the ranking of EXPONE over NO SHIFT forces Postposing (80c). When Postposing cannot occur, at last, the ranking of the item-specific DEP constraint over EXPONE forces the suspension of exponence in (80d).

(80) *Reinforcer Deletion*



7 Conclusion

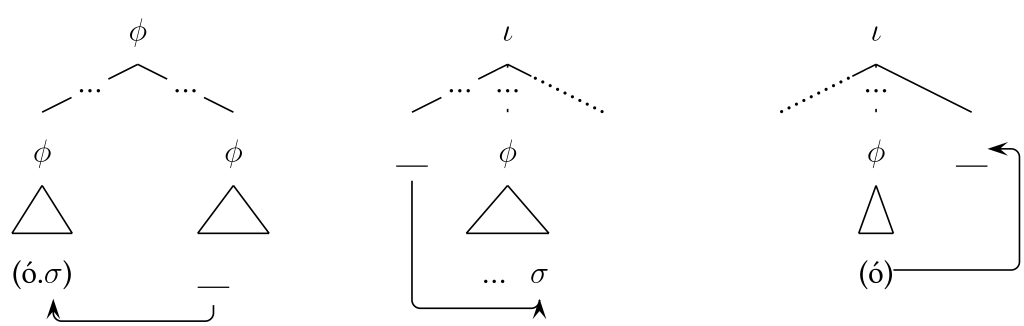
The result is a case that the systems of linearization and exponence must be split into two distinct stages and distributed across the morphological module and the phonology.

Empirically, this conclusion allows us to reinforce and extend the results of the wide literature on output-optimizing patterns of phonological displacement and exponence. Theoretically, in turn, it lends support to the hypothesis that the representations of the morphosyntax are translated into phonological terms in through an output-optimizing and imperfect translation. Conceptually, at last, they suggest that we might extend a similar treatment to other phenomena subsumed under the label of "PF effects"—and revisit the wider balance of labor between the morphology and the phonology.

To conclude our investigation, I would like to work back to the theory of movement itself. From the theoretical perspective developed here, it seems clear that the phonology can force relinearization in response to pressures of three types: constraints that force movement into particular positions, like the pressures behind Syllable Attraction (81a), constraints that force movement out of particular positions, like the pressures that shunt minimal elements into second position (81b), and pressures internal to moving elements themselves, like the tension that forces the reinforcers to postpose (81c).

(81) *Phonological Relinearization: Three Types*

a. *Attraction* b. *Push* c. *Greed*



The result is that there are phonological counterparts to all three forces that have been argued to drive displacement in the syntax: ATTRACT (Chomsky, 2001), PUSH constraints (Diesing, 1992; Moro, 2000; Stroik, 2009; Chomsky, 2013), and GREED (Chomsky, 1995; Bošković, 1995). Future work will tell what this connection may reveal.

References

Adger, David. 2006. Post-syntactic movement and the Old Irish verb. *Natural Language & Linguistic Theory*, **24**(3), 605–654.

Agbayani, Brian, & Golston, Chris. 2010. Phonological movement in classical Greek. *Language*, **86**(1), 133–167.

Agbayani, Brian, & Golston, Chris. 2016. Phonological constituents and their movement in Latin. *Phonology*, **33**(1), 1–42.

Agbayani, Brian, Golston, Chris, & Ishii, Toru. 2015. Syntactic and prosodic scrambling in Japanese. *Natural Language & Linguistic Theory*, **33**(1), 47–77.

Aissen, Judith. 2017. Special clitics and the right periphery in Tsotsil. *On looking into words (and beyond)*, **235**.

Anderson, Stephen R. 2008. English reduced auxiliaries really are simple clitics. *Lingue e linguaggio*, **7**(2), 169–186.

Aronoff, Mark, Arsyad, Azhar, Basri, Hasan, & Broselow, Ellen. 1987. Tier configuration in Makassarese reduplication. *Pages 1–15 of:* Bosch, A., Need, B., & Schiller, E. (eds), *Papers from the 23rd Annual Regional Meeting of the Chicago Linguistic Society, Part II: Parasession on Autosegmental and Metrical Phonology.* Chicago, IL: University of Chicago.

Arregi, Karlos, & Nevins, Andrew. 2012. *Morphotactics: Basque auxiliaries and the structure of spellout.* Vol. 86. Springer Science & Business Media.

Aylett, Matthew, & Turk, Alice. 2004. The smooth signal redundancy hypothesis: A functional explanation for relationships between redundancy, prosodic prominence, and duration in spontaneous speech. *Language and speech*, **47**(1), 31–56.

Basri, Hasan. 1999. *Phonological and syntactic reflections of the morphological structure of Selayarese.* Ph.D. thesis, State University of New York at Stony Brook.

Basri, Hasan, Broselow, Ellen, & Finer, Daniel. 1999. Clitics and crisp edges in Makassarese. *Toronto Working Papers in Linguistics*, **16**(2).

Basri, Hasan, Broselow, Ellen, & Finer, Daniel. 2012. The end of the word in Makassar languages. *Pages 111–147 of:* Borowsky, Toni, Kawahara, Shigeto, Shinya, Takahito, & Sugahara, Mariko (eds), *Prosody matters: Essays in honor of Lisa Selkirk.* Sheffield: Equinox.

Béjar, Susana. 1999. Agreement alternations and functional licensing in Selayarese. *Toronto Working Papers in Linguistics*, **16**(2).

Bennett, Ryan, & Elfner, Emily. 2019. The syntax–prosody interface. *Annual Review of Linguistics*, **5**, 151–171.

Bennett, Ryan, Elfner, Emily, & McCloskey, James. 2016. Lightest to the right: An apparently anomalous displacement in Irish. *Linguistic Inquiry*, **47**(2), 169–234.

Bernstein, Judy B. 1997. Demonstratives and reinforcers in Romance and Germanic languages. *Lingua*, **102**(2-3), 87–113.

Bobaljik, Jonathan David. 2000. The ins and outs of contextual allomorphy. *University of Maryland working papers in linguistics*, **10**, 35–71.

Bobaljik, Jonathan David. 2012. *Universals in comparative morphology: Suppletion, superlatives, and the structure of words.* MIT Press.

Booij, Geert. 1998. Phonological output constraints in morphology. *Phonology and morphology of the Germanic languages*, **143163**.

Booij, Geert. 1999. *The phonology of Dutch.* Oxford University Press.

Bošković, Željko. 1995. Case properties of clauses and the Greed Principle. *Studia Linguistica*, **49**(1), 32–53.

Brodkin, Dan. 2025a. The Prosody of the Extended VP. *Natural Language and Linguistic Theory.*

Brodkin, Dan. 2025b. Suppletion in Global Perspective. *Linguistic Inquiry.*

Brodkin, Dan. 2026. Demonstratives, Reinforcers, and Rightward Movement: Phonological Displacement in Mandar. *Linguistic Inquiry.*

Bye, Patrik, & Svenonius, Peter. 2012. Non-concatenative morphology as epiphenomenon1. *The morphology and phonology of exponence*, 427–495.

Chen, Matthew Y. 1987. The syntax of Xiamen tone sandhi. *Phonology*, **4**, 109–149.

Chomsky, Noam. 1995. *The Minimalist Program.* MIT press.

Chomsky, Noam. 2001. Derivation by phase. *Pages 1–52 of*: Kenstowicz, M. (ed), *Ken Hale: a life in language.* MIT Press.

Chomsky, Noam. 2013. Problems of projection. *Lingua*, **130**, 33–49.

Clemens, Lauren. 2019. Prosodic noun incorporation: The relationship between prosody and argument structure in Niuean. *Syntax*, **22**(4), 337–377.

Clemens, Lauren, & Coon, Jessica. 2018. Deriving verb-initial word order in Mayan. *Language*, **94**(2), 237–280.

De Belder, Marijke. 2020. A split approach to the selection of allomorphs: Vowel length alternating allomorphy in Dutch. *Glossa: a journal of general linguistics*, **5**(1).

Diesing, Molly. 1992. *Indefinites.* Cambridge, MA: MIT press.

Diesing, Molly, & Zec, Draga. 2011. Interface effects: Serbian clitics. *Experiments at the interfaces, syntax and semantics*, **37**, 1–30.

Diesing, Molly, & Zec, Draga. 2017. Getting in the first word: Prosody and predicate initial sentences in Serbian. *Glossa: a journal of general linguistics*, **2**(1).

Elfner, Emily. 2015. Recursion in prosodic phrasing: Evidence from Connemara Irish. *Natural Language & Linguistic Theory*, **33**(4), 1169–1208.

Elfner, Emily Jane. 2012. *Syntax-prosody interactions in Irish.* University of Massachusetts Amherst.

Elordieta, Gorka. 2006. Binarity at the left edge of Intonational Phrases. *In: GLOW Workshop on Prosodic Phrasing.*

Elordieta, Gorka. 2015. Recursive phonological phrasing in Basque. *Phonology*, **32**(1), 49–78.

Embick, David. 2000. Features, syntax, and categories in the Latin perfect. *Linguistic inquiry*, **31**(2), 185–230.

Embick, David. 2007. Linearization and local dislocation: Derivational mechanics and interactions. *Linguistic analysis*, **33**(3-4), 303–336.

Embick, David. 2010. *Localism versus globalism in morphology and phonology.* Vol. 60. mit Press.

Embick, David, & Noyer, Rolf. 2001. Movement operations after syntax. *Linguistic inquiry*, **32**(4), 555–595.

Esser, S.J. 1938. *Atlas van tropisch Nederland.* Amsterdam: Nederlandsch Aardrijkskundig Genootschap.

Féry, Caroline. 2013. Focus as prosodic alignment. *Natural Language & Linguistic Theory*, **31**(3), 683–734.

Féry, Caroline. 2017. *Intonation and prosodic structure.* Cambridge University Press.

Féry, Caroline, & Ishihara, Shinichiro (eds). 2016. *The Oxford handbook of information structure.* Oxford University Press.

Finer, Daniel. 1999. Cyclic clitics in Selayarese. *Toronto Working Papers in Linguistics*, **16**(2).

Finer, Daniel L. 1997. Contrasting Ā-dependencies in Selayarese. *Natural Language & Linguistic Theory*, **15**(4), 677–728.

Finer, Daniel L. 1998. Sulawesi relatives, V-raising, and the CP-complement hypothesis. *Canadian Journal of Linguistics/Revue canadienne de linguistique*, **43**(3-4), 283–306.

Friberg, Barbara. 1991. Ergativity, focus and verb morphology in several South Sulawesi languages. *Vical*, **2**, 103–130.

Friberg, Barbara. 1996. Konjo's peripatetic person markers. *Pacific Linguistics. Series A. Occasional Papers*, **84**, 137–171.

Friberg, Barbara, & Jerniati. 2000. *Mandar Conversations.* Unpublished; circulated by the Language Office of South Sulawesi (Balai Bahasa Sulawesi Selatan).

Friberg, Timothy, & Friberg, Barbara. 1991. Notes on Konjo phonology. *Studies in Sulawesi linguistics, part II*, 71–115.

Gafos, Adamantios, & Lombardi, Linda. 1999. Consonant transparency and vowel echo. *Page 8 of: North East Linguistics Society*, vol. 29.

Ghini, Mirco. 1993. Phi-formation in Italian: A new proposal. *Toronto working papers in linguistics*, **12**(2).

Göbbel, Edward. 2007. Extraposition as PF movement. *Proceedings of WECOL 2006*, 132–145.

Gouskova, Maria. 2012. Unexceptional segments. *Natural Language & Linguistic Theory*, **30**(1), 79–133.

Gouskova, Maria, & Bobaljik, Jonathan D. 2020. Allomorphy and vocabulary insertion. *Submitted to Handbook of Distributed Morphology, CUP.*

Grimes, Charles E, & Grimes, Barbara D. 1987. *Languages of south Sulawesi.* Dept. of Linguistics, Research School of Pacific Studies, The Australian National University.

Hale, Kenneth, & Selkirk, Elisabeth. 1987. Government and tonal phrasing in Papago. *Phonology*, **4**, 151–183.

Hall, T Alan. 1999. Phonotactics and the prosodic structure of German function words. *Amsterdam Studies in the Theory and History of Linguistic Science, Series IV*, 99–132.

Halle, Morris, & Marantz, Alec. 1993. Distributed morphology and the pieces of inflection. *Pages 111–176 of*: Hale, K., & Keyser, S. J. (eds), *The View from Building 20.* Cambridge, MA: MIT Press.

Halpern, Aaron. 1995. *On the placement and morphology of clitics.* Center for the Study of Language (CSLI).

Halpern, Aaron Lars. 1992. *Topics in the placement and morphology of clitics.* Ph.D. thesis, Stanford University.

Hamlaoui, Fatima, & Szendrői, Kriszta. 2015. A flexible approach to the mapping of intonational phrases. *Phonology*, **32**(1), 79–110.

Hamlaoui, Fatima, & Szendrői, Kriszta. 2017. The syntax-phonology mapping of intonational phrases in complex sentences: A flexible approach. *Glossa: a journal of general linguistics*, **2**(1).

Hargus, Sharon, & Tuttle, Siri G. 1997. Augmentation as affixation in Athabaskan languages. *Phonology*, **14**(2), 177–220.

Hayes, Bruce. 1981. *A metrical theory of stress rules.* Ph.D. thesis, MIT.

Hayes, Bruce. 1989. The prosodic hierarchy in meter. *Pages 201–260 of: Rhythm and meter.* Elsevier.

Hayes, Bruce, & Lahiri, Aditi. 1991. Bengali intonational phonology. *Natural language & linguistic theory*, **9**(1), 47–96.

Henderson, Robert. 2012. Morphological alternations at the intonational phrase edge. *Natural Language & Linguistic Theory*, **30**(3), 741–787.

Hewett, Matthew. 2023. Allomorphy in Semitic discontinuous agreement: Evidence for a modular approach to postsyntax. *Natural Language & Linguistic Theory*, **41**(3), 1091–1145.

Inkelas, Sharon, & Zec, Draga. 1990. *The phonology-syntax connection.* University of Chicago Press.

Inkelas, Sharon, & Zec, Draga. 1995. The phonology-syntax interface. *In:* Goldsmith, John A (ed), *Handbook of Phonological Theory.* Blackwell Publishing.

Itô, Junko, & Mester, Armin. 2007. Prosodic adjunction in Japanese compounds. *MIT working papers in linguistics*, **55**, 97–111.

Itô, Junko, & Mester, Armin. 2009. The extended prosodic word. *Phonological domains: Universals and deviations*, 135–194.

Itô, Junko, & Mester, Armin. 2012. Recursive prosodic phrasing in Japanese. *Prosody matters: Essays in honor of Elisabeth Selkirk*, 280–303.

Itô, Junko, & Mester, Armin. 2013. Prosodic subcategories in Japanese. *Lingua*, **124**, 20–40.

Itô, Junko, & Mester, Armin. 2019. Match as Syntax-Prosody MAX/DEP: Prosodic Enclisis in English. *English linguistics*, **36**(1).

Ito, Junko, & Mester, Armin. 2021. Recursive prosody and the prosodic form of compounds. *Languages*, **6**(2), 65.

Jukes, Anthony. 2006. *Makassarese (basa Mangkasara') A description of an Austronesian language of South Sulawesi.* Ph.D. thesis, The University of Melbourne.

Kager, René Willibrord Joseph. 1996. On affix allomorphy and syllable counting. *Studia grammatica*, **41**, 155–171.

Kaufman, Daniel. 2008. South Sulawesi pronominal clitics: Form, function and position. *Studies in Philippine languages and cultures*, **17**, 13–65.

Kawahara, Shigeto. 2007. Copying and spreading in phonological theory: Evidence from echo epenthesis. *UMOP*, **32**, 111–144.

King, Harold V. 1970. On blocking the rules for contraction in English. *Linguistic Inquiry*, **1**(1), 134–136.

King, Tracy Holloway. 1996. Slavic clitics, long head movement, and prosodic inversion. *Journal of Slavic linguistics*, 274–311.

Kramer, Ruth. 2010. The Amharic definite marker and the syntax–morphology interface. *Syntax*, **13**(3), 196–240.

Kubozono, Haruo. 1989. Syntactic and rhythmic effects on downstep in Japanese. *Phonology*, **6**(1), 39–67.

Kubozono, Haruo. 1993. *The Organization of Japanese Prosody.* Kurosio Publishers.

Kurisu, Kazutaka. 2001. *The phonology of morpheme realization.* Ph.D. thesis, University of California, Santa Cruz.

Kusmer, Leland. 2019. *Optimal linearization: Prosodic displacement in Khoekhoegowab and beyond.* Ph.D. thesis, University of Massachusetts, Amherst.

Kusmer, Leland Paul. 2020. Optimal Linearization: Word-Order Typology with Violable Constraints. *Syntax*, **23**(4), 313–346.

Ladd, D Robert. 1986. Intonational phrasing: the case for recursive prosodic structure. *Phonology*, **3**, 311–340.

Laskowske, Douglas. 2016. *Voice in Bugis: an RRG perspective.* MA Thesis, University of North Dakota.

Lee, Jason Kwok Loong. 2008. Transitivity, Valence and Voice in Mandar. *Studies in Philippine Languages and Cultures*, **19**, 55–66.

Liberman, Mark, & Prince, Alan. 1977. On stress and linguistic rhythm. *Linguistic inquiry*, **8**(2), 249–336.

López, Luis. 2009a. *A derivational syntax for information structure.* Vol. 23. Oxford University Press, USA.

López, Luis. 2009b. Ranking the linear correspondence axiom. *Linguistic Inquiry*, **40**(2), 239–276.

Mascaró, Joan. 1996. External allomorphy as emergence of the unmarked. *Page 473–483 of:* Durand, Jacques, & Laks, Bernard (eds), *Current trends in phonology: Models and methods*. Salford, Manchester: University of Salford, European Studies Research Institute. Reprinted in John J. McCarthy (ed.). 2004. Optimality Theory in Phonology: A reader, 513–522. Oxford: Blackwell.

Mascaró, Joan. 2007. External allomorphy and lexical representation. *Linguistic Inquiry*, **38**(4), 715–735.

Matti, David F. 1994. Mamasa pronoun sets. *Studies in Sulawesi Linguistics*, **3**, 65–89.

McCarthy, John J. 1998. Morpheme structure constraints and paradigm occultation. *Pages 123–150 of:* Gruber, M. Catherine, Higgins, Derrick, Olson, Kenneth, & Wysocki, Tamra (eds), *Proceedings of CLS 32 Part 2: The Panels*. Chicago, IL: Chicago Linguistic Society.

McCarthy, John J. 2011. Pausal phonology and morpheme realization. *Pages 341–373 of:* Borowsky, Toni, Kawahara, Shigeto, Shinya, Takahito, & Sugahara, Mariko (eds), *Prosody matters: Essays in honor of Lisa Selkirk*. Sheffield: Equinox.

McCarthy, John J, & Prince, Alan. 1993. Generalized alignment. *Pages 79–153 of: Yearbook of morphology 1993*. Springer.

Merchant, Jason, & Pavlou, Natalia. 2017. The morphosyntax of the periphrastic future under negation in Cypriot Greek. *Journal of Greek Linguistics*, **17**(2), 233–262.

Mester, R Armin. 1994. The quantitative trochee in Latin. *Natural Language & linguistic theory*, **12**(1), 1–61.

Mills, Roger F. 1975. *Proto South Sulawesi and Proto Austronesian Phonology*. Ph.D. thesis, University of Michigan.

Mithun, Marianne, & Basri, Hasan. 1986. The phonology of Selayarese. *Oceanic Linguistics*, **25**(1/2), 210–254.

Moro, Andrea. 2000. *Dynamic antisymmetry*. Cambridge, MA: MIT Press.

Muthalib, Abdul, & Sangi, M Zain. 1991. *Puisi Kalindaqdaq Mandar*. Departemen Pendidikan dan Kebudayaan.

Myler, Neil. 2013. Linearization and post-syntactic operations in the Quechua DP. *Pages 171–210 of: Challenges to linearization*. Mouton de Gruyter Berlin.

Myrberg, Sara. 2010. *The intonational phonology of Stockholm Swedish*. Ph.D. thesis, Acta Universitatis Stockholmiensis.

Myrberg, Sara. 2013. Sisterhood in prosodic branching. *Phonology*, **30**(1), 73–124.

Nespor, Marina, & Vogel, Irene. 1986. *Prosodic phonology*. De Gruyter Mouton.

Noyer, Rolf. 1994. Mobile affixes in Huave: Optimality and morphological wellformedness. *Pages 67–82 of: Proceedings of the Twelfth West Coast Conference on Formal Linguistics. Stanford: CSLI.*

Ostrove, Jason. 2018. Stretching, spanning, and linear adjacency in Vocabulary Insertion. *Natural Language & Linguistic Theory*, **36**(4), 1263–1289.

Pater, Joe. 1999. Austronesian nasal substitution and other *NC effects. *Pages 310–343 of:* Kager, René, van der Hulst, Harry, & Zonneveld, Wim (eds), *The prosody-morphology interface*. Cambridge University Press.

Pater, Joe. 2000. Non-uniformity in English secondary stress: the role of ranked and lexically specific constraints. *Phonology*, **17**(2), 237–274.

Pater, Joe. 2006. The locus of exceptionality: Morpheme-specific phonology as constraint indexation. *Page 259–296 of:* Bateman, Leah, O'Keefe, Michael, Reilly, Ehren, & Werle, Adam (eds), *Papers in Optimality Theory III*. Amherst, MA: Graduate Linguistics Student Association, University of Massachusetts.

Pater, Joe. 2008. Morpheme-specific phonology: Constraint indexation and inconsistency resolution. *In:* Parker, Steve (ed), *Phonological argumentation: Essays on evidence and motivation*. London: Equinox.

Pelenkahu, RA, Muthalib, Abdul, & Sangi, M Zain. 1983. *Struktur Bahasa Mandar*. Pusat Pembinaan dan Pengembangan Bahasa, Departemen Pendidikan dan Kebudayaan.

Peperkamp, Sharon Andrea. 1997. *Prosodic words*. Vol. 34. Holland Academic Graphics The Hague.

Perlmutter, David M. 1998. Interfaces: explanation of allomorphy and the architecture of grammars. *Pages 307–338 of:* Lapointe, Stephen, Brentari, Diane, & Farrel, Patrick (eds), *Morphology and its relation to phonology and syntax*. Palo Alto, CA: Stanford Center for the Study of Language and Information Publications.

Potsdam, Eric. 2022. Malagasy extraposition: Evidence for PF movement. *Natural Language & Linguistic Theory*, **40**(1), 195–237.

Prieto, Pilar. 2005. Syntactic and eurhythmic constraints on phrasing decisions in Catalan. *Studia Linguistica*, **59**(2-3), 194–222.

Prince, Alan, & Smolensky, Paul. 2004. *Optimality Theory: Constraint interaction in generative grammar*. John Wiley & Sons.

Raffelsiefen, Renate. 1996. Gaps in word formation. *Pages 194–209 of: Interfaces in phonology*. Akademie Verlag.

Raffelsiefen, Renate. 1999. *Phonological constraints on English word formation*. Springer.

Raffelsiefen, Renate. 2004. Absolute ill-formedness and other morphophonological effects. *Phonology*, **21**(1), 91–142.

Roehrs, Dorian. 2010. Demonstrative-reinforcer constructions. *The Journal of Comparative Germanic Linguistics*, **13**, 225–268.

Ross, John Robert. 1967. *Constraints on variables in syntax*. Ph.D. thesis, MIT.

Rubach, Jerzy, & Booij, Geert E. 2001. Allomorphy in optimality theory: Polish iotation. *Language*, 26–60.

Schütze, Carson. 1994a. Serbo-Croatian second position clitic placement and the phonology-syntax interface. *MIT working papers in linguistics*, **21**, 373–473.

Schütze, Carson. 1994b. Serbo-Croatian second position clitic placement and the phonology-syntax interface. *Pages 373–473 of:* Toman, Jindrich (ed), *Annual Workshop on Formal Approaches to Slavic Linguistics: The College Park Meeting*. Michigan Slavic Materials: Ann Arbor.

Selkirk, Elisabeth. 1984. *Phonology and syntax: the relation between sound and structure*. Ph.D. thesis, MIT.

Selkirk, Elisabeth. 1986. On derived domains in sentence phonology. *Phonology*, **3**, 371–405.

Selkirk, Elisabeth. 1995a. The prosodic structure of function words. *University of Massachusetts occasional papers in linguistics*, **18**, Urbanczyk.

Selkirk, Elisabeth. 1995b. Sentence prosody: Intonation, stress, and phrasing. *The handbook of phonological theory*, **1**, 550–569.

Selkirk, Elisabeth. 2009. On clause and intonational phrase in Japanese: The syntactic grounding of prosodic constituent structure. *Gengo Kenkyu*, **136**, 35–73.

Selkirk, Elisabeth. 2011. The syntax-phonology interface. *The handbook of phonological theory*, **2**, 435–483.

Selkirk, Elisabeth, & Elordieta, Gorka. 2010. The role for prosodic markedness constraints in phonological phrase formation in two pitch accent languages. *Tone and Intonation in Europe*, **4**, 9–11.

Shih, Shu-hao. 2016. Binarity and focus in prosodic phrasing: New evidence from Taiwan Mandarin. *In: Proceedings of the Annual Meetings on Phonology*.

Sikki, Muhammad, Muthalib, Abdul, Mulya, Abdul Kadir, & Haddade, Muhammad Naim. 1987. *Kata Tugas Bahasa Mandar*. Pusat Pembinaan dan Pengembangan Bahasa, Departemen Pendidikan dan Kebudayaan.

Snedeker, Jesse, & Trueswell, John. 2003. Using prosody to avoid ambiguity: Effects of speaker awareness and referential context. *Journal of Memory and language*, **48**(1), 103–130.

Stanton, Juliet, & Zukoff, Sam. 2018. Prosodic identity in copy epenthesis. *Natural Language & Linguistic Theory*, **36**(2), 637–684.

Stroik, Thomas. 2009. *Locality in minimalist syntax*. Cambridge, MA: MIT Press.

Strømme, Kari K. 1994. Person marking in the Mamuju language. *Studies in Sulawesi Linguistics*, **3**, 91–113.

Truckenbrodt, Hubert. 1995. *Phonological phrases–their relation to syntax, focus, and prominance*. Ph.D. thesis, Massachusetts Institute of Technology.

Truckenbrodt, Hubert. 1999. On the relation between syntactic phrases and phonological phrases. *Linguistic inquiry*, **30**(2), 219–255.

Truckenbrodt, Hubert. 2002. Variation in p-phrasing in Bengali. *Linguistic Variation Yearbook*, **2**(1), 259–303.

Truckenbrodt, Hubert. 2007. The syntax-phonology interface. *The Cambridge handbook of phonology*, **435456**.

Tyler, Matthew. 2019. Simplifying MATCH WORD: Evidence from English functional categories. *Glossa: a journal of general linguistics*, **4**(1).

Valkama, Kari. 1995a. Person marking in Duri. *Studies in Sulawesi linguistics*, **4**, 47–95.

Valkama, Susanne. 1995b. Notes on Duri transitivity. *Studies in Sulawesi Linguistics part IV: NUSA*, **37**, 2–45.

Van Handel, Nicholas. 2021. Matching overtly headed syntactic phrases in Italian. *Phonology*, **38**(2), 317–356.

Wagner, Michael. 2005. *Prosody and recursion*. Ph.D. thesis, Massachusetts Institute of Technology.

Wagner, Michael. 2010. Prosody and recursion in coordinate structures and beyond. *Natural Language & Linguistic Theory*, **28**(1), 183–237.

Watson, Duane G, Gunlogson, Christine A, & Tanenhaus, Michael K. 2006. Online methods for the investigation of prosody. *Methods in empirical prosody research*, 259–282.

Werle, Adam. 2009. *Word, phrase, and clitic prosody in Bosnian, Serbian, and Croatian.* Ph.D. thesis, University of Massachusetts Amherst.

Yu, Alan. 2007. *A natural history of infixation.* Vol. 15. OUP Oxford.

Zec, Draga. 2005. Prosodic differences among function words. *Phonology*, **22**(1), 77–112.

Notes

[1]GLOSSING: ABS: absolutive, ANTIP: antipassive, ERG: ergative, GEN: genitive, PASS: passive.

[2]This requirement persists beyond the system of deixis. Mandar requires the demonstrative *do* to surface above many types of nominals that are discourse-anaphoric, and in this context, it continues to require *o*, even though it is semantically anomalous to introduce regular locative adverbs into DPs of this type (82a)-(82b). The regular locative adverbs are never required in any corresponding way: they are always optional and they can always co-occur with minimally mismatched" demonstratives, such that *indi* and *dini* can surface beneath *ndi* and *de* and *dio* and *diting* can surface beneath *do* and *iting* (82c)-(82d).

(82) *Selection Beyond Deixis*

a. Diang da'dua analisa: mesa sittassis, mesa ponologi
there are two analysis one syntactic one phonological
'There are two analyses: one syntactic, one phonological.'

b. Macoa i [$_{DP}$ do analisa ponologi (#dio) *(o)].
good 3ABS that analysis phonological there there
'The phonological analysis (#there) is elegant.'

c. De posa [**indi** / **dini** / ___] e
this cat here here here
'This cat here'

d. Do posa [**dio** / **diting** / ___] o
that cat there there there
'That cat there'

[3]In the investigation that follows, we will restrict our attention to the type of prosody that emerges most readily when speakers are asked to introspect on the most natural pronunciation of particular strings. Following the methodology of Nespor & Vogel 1986 and my prior work on Mandar (Brodkin, 2025a), I will thus abstract away from the many factors that can derail patterns of prosodic phrasing in the context of natural speech (Snedeker & Trueswell, 2003; Aylett & Turk, 2004; Watson *et al.*, 2006; Féry & Ishihara, 2016).

[4]Many other systems in the phonology of the language converge with these two effects to single out the same domain: for instance, the edges that block Nasal Assimilation and host Comma Intonation also (*i*) host reinforcers (this section), (*ii*) license the emergence of special types of feet (Section 5); (*iii*) license the emergence of particular glottal segments; (*iv*) host the types of intonational contours canonically associated with the ι, and (*v*) host a series of "final particles" that descriptively associate with contours of this type.

[5]For discussion of several other cases of putative phonological displacement, which targets elements and landing sites that seem more easily defined in syntactic terms, see Göbbel 2007; López 2009a; Werle 2009; Agbayani & Golston 2010; Agbayani *et al.* 2015; Agbayani & Golston 2016; Clemens & Coon 2018; Clemens 2019; Potsdam 2022.

[6]This description breaks from the traditional analysis of word-level stress in the language, which takes it to fall on the penult without regard to syllable weight. This quantity-insensitive analysis has been asserted and advanced in the grammar of Mandar by R.A Pelenkahu, Abdul Muthalib, and Zain Sangi (Pelenkahu *et al.*, 1983), in several of my own works (Brodkin 2025b) and in every discussion of stress that I am aware of in every other language of South Sulawesi (Friberg & Friberg 1991; Jukes 2006, and especially Mithun & Basri 1986; Basri *et al.* 1999; Basri 1999; Basri *et al.* 2012, a series of works on Selayarese that are co-authored by Hasan Basri, a native speaker of that language). Over my first seven years of hearing, speaking, and working on Mandar, I assumed this analysis erroneously for six reasons: (*i*) I was unwilling to make judgments about metrical structure on the basis of my impressions about phonetic prominence (though the rightmost trochee in the word clearly hosts word-level stress (Pelenkahu *et al.*, 1983) and I believe that an audible secondary stress falls on the head of each trochee to the left); (*ii*) word-final monosyllabic feet are ruled out in many positions by the Rhythm Rule (Footnote X) and the Positional Branching Constraint (this subsection), (*iii*) word-final monosyllabic feet are favored at the right edge of the intonational phrase (next subsection), (*iv*) the right edge of the intonational phrase hosts so many independent phonetic distractions and higher-level phonological events that it is difficult to single out word-level prominence in that position, (*v*) the system of footing is derailed in complex ways by the phonology of focus, which interacts with every other system in the phrasal phonology as well, and (*vi*) the interactions between phonetic prominence, word-level metrical structure, and top-down constraints are so complex in Indonesian, the language of contact for this work, that I have been slow to interpret my own intuitions about Mandar.

[7]This effect is segmentally similar to the "echo epenthesis" of the related Makassarese, Selayarese, and Konjo, which applies in a different prosodic context; for discussion, see (Aronoff *et al.*, 1987; Friberg & Friberg, 1991; McCarthy, 1998; Gafos & Lombardi, 1999; Basri *et al.*, 1999; Kawahara, 2007; Stanton & Zukoff,

2018).

[8]We can see the same effect across all of the classes of monosyllabic heads studied in Brodkin 2025b. The native adverbs *bo* 'again' and *to* 'also" to begin, are monosyllabic in phrase-medial positions, disyllabic at the right edges of medial phonological phrases that are final in the maximal phonological phrase (83a), and monosyllabic again at the right edge of the intonational phrase (83b).

(83) *Native Monosyllables: Epenthesis Suspended*

a.

bukúna^{H}	wémme	**(wóʔo)**^{H}	maníni^{H}.
buku-na	bemmeʔ	bo	manini
book-3GEN	fall	again	later

'The book will fall again later.'

b.

bukúna^{H}	wémme	**(wóʔ)**^{H}.
buku-na	bemmeʔ	bo
book-3GEN	fall	again

'The book will fall again.'

The same effects can be seen with the preposition *sung* 'out' and the negator *da* 'don't.' Like all functional heads in the language, these elements are typically parsed as stray syllables and directly integrated into the higher prosodic domains built around their complements (and see also Selkirk 1995a; Peperkamp 1997; Booij 1999; Hall 1999; Itô & Mester 2009; Tyler 2019). When stranded by complement ellipsis, they are forced to fall at the edges of phonological phrases that correspond to the projections they head (and see Anderson 2008; Itô & Mester 2019; Brodkin 2025b, as well as King 1970). In this position, they undergo ʔv-epenthesis at the right edges of medial phonological phrases that are final in the maximal phonological phrase but cease to undergo this repair at the edge of the intonational phrase.

(84) *Monosyllabic Functional Heads: Parallel Effects*

a.

bémme^{H}		**súʔun**^{H}		diɣéna?.
bemmeʔ	i	suŋ	___	digenaʔ
fall	3ABS	out	NP	earlier

'It fell out earlier.'

b.

bémme^{H}		**(súŋ)**^{H}.
bemmeʔ	i	suŋ
fall	3ABS	out

'It fell out.'

[9]These results turn on two further hypotheses about the inventory of possible constraints. First, I assume that the interface cannot detect the boundaries of syntactic islands in the morphological representations it receives. Second, I assume that the interface cannot leverage C-command relationships in the input to restrict movement to phonologically defined positions. These hypotheses rule out two types of constraints: those that would that prohibit phonological relinearization out of syntactic islands and those that would force phonological relinearization to select its targets on the basis of relative C-command.